

The Causal Effect of Violent Crime on Labor Market: Evidence from a Ceasefire

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Abstract

While violence and under-development often coexist, the causal impact of reduced violence on labor markets remains poorly understood. I exploit El Salvador's 2012 gang truce, a sharp and plausibly exogenous 60% drop in homicides, to provide causal evidence that falling violence yields sizable labor market gains. Using administrative data on firms and crime, I show that a one-standard-deviation decline in violence raises employment by 8.4 percent and active firms by 4.6 percent within a year. Evidence suggests these gains arise from improved consumer confidence, which boosts local spending, reduces firm exits, and increases labor demand, especially in non-tradable sectors.

Keywords: Organized crime, labor markets, violence reduction, economic impact, business sustainability, economic growth

JEL codes: D74, K42, O17

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1 Introduction

A central question in economics is how violent crime affects economic activity. Despite its relevance across both developed and developing countries, causal evidence remains scarce, largely because sudden and plausibly exogenous changes in violence are rare. This gap in the literature is striking given that criminal organizations and violent crime remain widespread. In Latin America and the Caribbean (LAC), for example, homicide rates exceed the global average by more than threefold, and organized criminal groups account for roughly one-half of all killings (United Nations Office on Drugs and Crime (UNODC), 2023). Moreover, 54% of households report the presence of criminal organizations in their neighborhoods (Uribe et al., 2022), raising important questions about their consequences for local economic performance. While much research explores how economic improvements can curb criminal activity,¹ the reverse channel, how violent crime itself suppresses economic performance, remains poorly understood.

This paper provides causal evidence on the economic impact of violent crime by exploiting a uniquely sharp and unexpected reduction in homicides during the 2012 Salvadoran gang truce. Secretly negotiated between incarcerated gang leaders and government officials, the agreement cut homicide rates by an estimated 60% within weeks. The episode offers an unusually natural experiment to isolate how enhanced security affects labor markets and firm activity.

Exploiting the truce’s sharp geographic and temporal variation, comparing areas heavily exposed to the ceasefire with other units that were not, before and after its onset, I document sizable gains in local labor markets. A one-standard-deviation decline in homicides raises employment by 8.4% and the number of firms by 4.6% within the first year. These effects appear to be driven by improvements in consumer confidence: as local residents perceived greater safety, they reengaged with neighborhood markets, boosting demand in local (non-tradable) sectors. In turn, firms expanded operations, faced fewer closures, and increased hiring, leading to higher wages. This interpretation is consistent with the fact that the economic gains are concentrated in non-tradable sectors and are accompanied by increased sales, fewer business closures, rising wages, and improved consumer confidence. Finally, positive spillovers to adjacent areas suggest that these security gains generated

¹For early analyses, see Jones (1932) and Simpson (1932), as well as more comprehensive discussions by Bushway and Reuter (2011) and Mustard (2010). Specific economic shocks studied include oil shocks (Raphael and Winter-Ebmer, 2001), changes in industrial composition through Bartik instruments (Gould, Weinberg and Mustard, 2002; Fougère, Kramarz and Pouget, 2009), exchange rate movements (Lin, 2008), international mineral prices (Axbard, Benshaul-Tolonen and Poulsen, 2019), trade liberalization (Dix-Carneiro, Soares and Ulyssea, 2016), mass layoffs and plant closures (Bennett and Ouazad, 2020; Pinotti, Britto and Sampaio, 2020), and improvements in legal market conditions (Pinotti, 2017).

broader economic benefits beyond directly treated zones.

The temporary nature of the truce highlights the fragility of these gains. When violence resurged in 2014, the number of active firms reverted to its pre-truce level, meaning that the earlier increase in the stock of firms was essentially wiped out. Instead, employment ceased to grow relative to the counterfactual and resumed its earlier trajectory, suggesting that while labor markets retained some of their gains, the underlying growth potential was not sustained. This divergence underscores that preserving economic improvements requires lasting rather than temporary security interventions. Consistent with these local effects, national statistics also show a temporary rise in aggregate labor force participation during the truce years, a pattern not observed in other countries in the region, that faded once violence resurged.

To identify the effects of crime reduction on economic activity, I employ a difference-in-differences (DiD) framework. Treatment and control units are classified based on sharp declines in homicides that occurred shortly after the onset of the gang truce and stood out relative to each area's historical crime trends. Units experiencing sharp and unexpected drops in violence are more plausibly linked to gang negotiations. This classification strategy provides a higher-powered approach, as the sudden and pronounced reductions likely dominate potential biases. To address remaining concerns, I conduct placebo tests, replicate the analysis using an alternative classification based on pre-truce characteristics—such as prior gang-related activity—and perform a series of sensitivity checks. Building on the same classification, I also construct a continuous measure of exposure to homicide reductions, which allows me to examine the relationship between crime declines and economic outcomes, estimate treatment intensity, and test for linear and nonlinear functional forms.

The estimates of this paper likely understate the economic benefits of crime reduction. Because the truce was temporary and its credibility uncertain, firms and workers may have postponed investments and hiring that would have occurred under a permanent decline in violence. In addition, the agreement primarily reduced homicides; other offenses such as extortion showed no clear decline, and evidence from a later non-aggression pact among the same gangs suggests that extortion may even rise in some contexts.² A more stable and broad-reaching reduction in crime would therefore be expected to generate larger increases in economic activity than those estimated here.

This paper contributes to the growing literature on how organized crime and violence shape economic outcomes, particularly firm performance and labor markets. Previous studies document

²Brown et al. (2024) studies a 2016 non-aggression pact and documents higher reported extortion, drawing on payment records from a logistics company that distributes goods across El Salvador.

that criminal activities and civil conflicts negatively impact economic indicators such as stock market returns (Abadie and Gardeazabal, 2003, 2019; Guidolin and La Ferrara, 2007), the performance of export-oriented firms (Ksoll, Macchiavello and Morjaria, 2010), firm survival probabilities (Camacho and Rodriguez, 2013; Collier and Hoeffler, 2004), market prices (Rozo, 2018), and—through mafia or cartel infiltration of local governments and private firms—curtail employment growth and undermine financial stability (Fenizia and Saggio, 2024; Mirenda, Mocetti and Rizzica, 2022). On the other hand, some studies have attempted to quantify the economic benefits of crime reduction through perceived valuations (Ludwig and Cook, 2001; Cook, Jeuland and Ludwig, 2025).

Closer to the question examined here, some studies have specifically analyzed how violence affects labor market participation. For instance, Utar (2024) and Velásquez (2020) document that increased violence discourages labor market participation due to heightened fears of victimization, notably among self-employed women who shift labor hours toward household activities, and blue-collar workers in labor-intensive manufacturing firms.

This paper departs from and expands upon this literature by leveraging a uniquely sharp and plausibly exogenous reduction in violent crime to provide direct causal evidence of its impact on economic activity. To my knowledge, it is among the first studies to show that a massive and unexpected decline in homicides can trigger substantial increases in employment and firm creation. The speed and magnitude of the response stand in contrast to prior work, which has typically examined gradual policy changes. For example, existing studies have focused on the multi-year escalation of Mexico’s drug war (2007–2011) or on Colombia’s more gradual homicide decline from the mid-1990s onward. In contrast, the Salvadoran truce produced a sudden and sustained drop in homicides that was both unexpected and unusually sharp, offering an ideal setting for identification by minimizing the scope for confounding dynamics, such as economic growth gradually improving security, that often complicate causal interpretation.

In addition to exploiting this distinctive shock, the analysis covers the full spectrum of economic activity rather than subsets of firms with high productivity or international exposure. Prior studies typically focus on stock market-listed firms, exporters, or specific manufacturing industries, while this paper examines broad effects across all sectors. A novel finding is that smaller firms and those in non-tradable sectors, more dependent on local demand, experience the most significant gains. This result complements recent evidence by Utar (2024), who shows that within manufacturing, plants that rely more heavily on domestic inputs and markets are especially vulnerable to violence. While that study is limited to one sector, my findings demonstrate that the economic benefits of

improved security extend well beyond manufacturing and are structurally driven by the importance of local demand of goods across multiple sectors. This has broader implications for understanding how violence distorts economic geography in high-crime contexts.

Earlier work often shows that increases in crime reduce labor supply, leading to lower employment and upward pressure on wages, consistent with a leftward shift in labor supply. By contrast, in this setting both employment and wages rose when violence fell, pointing to a demand-side mechanism. To interpret these patterns, I develop a simple theoretical model that treats crime as a consumer “fear cost” that raises the effective price in violent areas and includes a crime-sensitive participation margin (safer conditions shift the labor-supply schedule outward at a given wage). The model replicates the direction of the effects seen in the data and clarifies the conditions under which wages increase when violence falls—namely, when the demand expansion in formerly violent markets dominates the outward supply shift. A simple implication follows: the aggregate response hinges on where safety improves. When the crime-affected segment of the economy is large—as in El Salvador—the decline in fear costs can generate a sizable consumer-side expansion, making the demand channel the natural driver of the joint rise in wages and employment.

An additional question is whether the effects of crime reduction extend beyond directly treated areas. Spillovers matter for two reasons: they may bias the main estimates if neighboring areas are indirectly affected, and they are also outcomes of independent interest. To address this, I examine whether safer conditions in treated locations affected adjacent areas. While much of the prior literature highlights negative spillovers—crime being displaced into nearby neighborhoods (Draca, Machin and Witt, 2011; Dell, 2015)—I instead find suggestive evidence of positive spillovers, with reductions in violence fostering economic gains beyond the directly exposed units. Although the evidence is suggestive and subject to some classification limitations, the exercise indicates that the overall results remain robust: estimates of employment and firm creation stay positive even when spillovers are taken into account.

Although the institutional details of the Salvadoran truce are unique, the conditions under which its effects emerged, such as high fear costs, the importance of violent areas in overall consumption, and the reliance of households and firms on local demand, are not specific to El Salvador. Similar environments exist in other countries, suggesting that the mechanisms identified here are plausibly relevant beyond this case. At the same time, prior work has documented broader associations between crime and economic performance. For instance, Islam (2014) show that economic activity is negatively correlated with crime across firms in 27 developing countries, with the relationship

especially pronounced for small and medium firms. By moving from such correlations to causal evidence based on a sharp and unexpected decline in violence, this paper contributes to a broader understanding of how reducing crime can foster economic growth, and under what circumstances such gains are most likely to materialize.

The remainder of the paper is structured as follows. Section 2 provides background on violence in El Salvador and the 2012 gang truce. Section 3 describes the unique dataset constructed for this analysis, combining administrative and crime data. Section 4 outlines the identification strategy, including the difference-in-differences framework and robustness checks. Section 5 presents the main findings on the impacts of crime reduction on employment and firm activity. Section 6 explores the mechanisms driving these effects, focusing on reduced firm closures, improved sales, and sector-specific dynamics. Section 7 examines spillover effects, and Section 8 analyzes the economic reversal following the truce's collapse. Section 9 discusses the implications of the theoretical model and how it helps interpret the empirical results. Section 10 concludes.

2 Context

2.1 Violence in El Salvador

El Salvador has previously been characterized by high levels of violence often driven by gang activities. The country's experience with violent crime, particularly homicides and extortion, was heavily linked to the activities of gangs such as MS-13 and Barrio 18. These gangs, formed by Latin American migrants in Los Angeles, grew in strength and number upon their return to El Salvador, exacerbated by weak institutions and lax law enforcement (Arana, 2005). By 2010, crime had become the country's most pressing issue, with 61% of Salvadorans identifying it as their top concern, surpassing economic issues (35%) (Lab, 2010).

The arrival of deported gang members from the US in the mid-1990s led to the replication of behaviors and structures familiar in the contentious environments of Los Angeles. These individuals formed local chapters of MS-13 and Barrio 18, bringing with them gang symbols, language, and tattoos, and creating a collective identity through violence. Further, this period was characterized by a lack of government programs to address gang violence or reintegrate deported Salvadorans, which enabled a further escalation of the gang crisis.

Data from the National Police indicate that at least one-third of homicides are attributable to gang-related activities. The economic ramifications of this violence are substantial, with estimates

suggesting that it accounts for between 6.5% to 16% of the country's Gross Domestic Product (GDP) (Jaitman et al., 2017; Peñate et al., 2016). Crime has also been a persistent barrier to investment, with 47% of businesses identifying it as the most critical factor affecting their investment decisions in El Salvador in 2010 (FUSADES, 2010–2020). Studies by Melnikov, Schmidt-Padilla and Sviatschi (2020); Kalsi (2018); Castro et al. (2025); Castro and Kotti (2022); Castro et al. (2019) reveal the profound impact of gang presence on a wide spectrum of socio-economic factors within these neighborhoods. Notably, households in these areas face significant challenges concerning income, educational opportunities, housing quality, and electoral participation, all of which are negatively influenced by the pervasive control exerted by gangs.

2.2 Truce Negotiations

In 2012, a pivotal change occurred when the Salvadoran government facilitated a truce between MS-13 and Barrio 18. This truce, mediated with the support of the Catholic Church, led to a significant drop in homicides. However, the government's role in these negotiations remained controversial, with official denials of direct involvement despite apparent concessions to gang leaders. Moreover, not all gangs operating in El Salvador were part of the truce (Insight Crime, 2015).

This truce proved remarkably effective in curbing the nation's elevated homicide rates, achieving a roughly 60% reduction in such incidents—a decline unparalleled in the country's recent history. During this period, El Salvador recorded its lowest homicide levels since the end of the civil war, marking an unprecedented episode of relative peace. As part of the agreement, the government conceded to improve prison conditions for incarcerated gang leaders.

While the truce was successful in reducing homicides, the number of reported extortions remained nearly unchanged compared to pre-truce levels (see Figure B.5). The truce lasted approximately 24 months. Following this period, the Minister of Security, who was identified as the facilitator of the truce, was dismissed, gang leaders were deprived of the agreed privileges, and a new security plan was initiated. This new strategy adopted a far more confrontational approach toward gangs, intensifying direct clashes between security forces and gang members. Consequently, crime rates surged beyond pre-truce levels, culminating in a homicide rate of 103 per 100,000 inhabitants in 2015, the highest recorded since the civil war.

Beyond the immediate impact on violence, the truce was also accompanied by an expansion in labor force participation at the national level. As reported in Figure B.6, participation increased during the truce, but this gain was fully reversed once the truce collapsed. Importantly, such fluctu-

ations are not observed in other Latin American countries during the same period, highlighting the unique trajectory of El Salvador during this period.

3 Data

This study relies on three primary data sources, which are integrated and aggregated at the geographic level. Specifically, the unit of analysis is a set of 1,581 polygonal geographic divisions level units—covering the entire country. All firm and crime data are aggregated to these polygons and observed over time, yielding a monthly panel from 2010 to 2015.

1. **Social Security Database (ISSS):** This administrative dataset provides monthly records of all private-sector firms registered with the Salvadoran Social Security Institute. It includes information on firms' primary activity, number of employees, total payroll.
2. **National Civil Police Records:** This dataset offers monthly records of all reported homicides in the country.
3. **Annual Business Surveys from FUSADES:** Conducted by a leading Salvadoran think tank, these surveys cover approximately 580 firms from 2008 to 2015 and include self-reported indicators such as sales, perceived insecurity, and business expectations.

The social security and crime records were combined to construct a balanced monthly panel from 2010 to 2015, encompassing 1,581 geographic polygons throughout the country. The FUSADES survey data, available at an annual frequency, provide a complementary layer of firm-level information that can be linked to these polygons.

One potential limitation of the data is that firm records from the Social Security Institute (ISSS) include only a single registered address per company. For large firms with multiple branches or operational sites, this could mean I observe only their headquarters or administrative address, rather than their full geographic footprint. As a result, interpreting local effects for these firms is less straightforward. To address this, the main analyses focus on firms in the bottom 95% of the employment distribution, where the registered location is more likely to coincide with actual operations. Nevertheless, I also report results that include only the top 5% of firms and the full sample, ensuring that the findings are not driven by sample selection or omitted heterogeneity in firm size.

3.1 Identifying Treated Units

Precisely measuring gang presence poses significant challenges, as direct indicators are often unreliable or unavailable. To address this, I rely on a proxy measure that captures unusually large reductions in homicide rates shortly after the onset of the truce. The logic is that areas most influenced by the agreement will display a marked and abnormal drop in violent crime during this initial period, indicative of a sudden shift in gang behavior and diminished violent activity.

For the main analysis, I classify polygons as “treated” if they fall within the top 50% of the distribution of homicide reductions ($Z_{i,T}$) during the truce period (T) (i.e., those experiencing the most abnormally large homicide reductions relative to their historical variation). Robustness checks using alternative thresholds (such as the top 40%, 30%, 20%, or 10%) suggest consistent results: polygons experiencing more pronounced drops in homicides also exhibit stronger improvements in local labor market outcomes, regardless of the specific cutoff chosen. This pattern further supports the identification strategy, as increasingly restrictive thresholds should more precisely capture areas that experienced an exogenous shock due to the truce. Further methodological details on the construction of this measure, including the historical normalization of homicide fluctuations, are provided in Appendix A.1.

3.1.1 Addressing Potential Endogeneity

A key concern in interpreting the results is whether any decline in crime could mechanically lead to improvements in employment and firm activity. If this were the case, part of the estimated effect might not stem from the exogenous shock of the truce but rather from a more endogenous relationship between crime reductions and economic growth. This would raise the possibility that my findings do not capture the causal impact driven by the truce or that at least a portion of the observed effects could be attributed to broader economic dynamics rather than the truce itself.

To address this concern, I implement multiple placebo exercises using a triple-differences framework, in which I compare estimated effects across multiple hypothetical truce periods. Specifically, I fully reapply the classification methodology to placebo “truce” events occurring several months before the actual start date. For each placebo scenario (e.g., 3, 5, 7, or 9 months before the truce), I:

1. Compute the historical distributions of homicide changes and construct the $Z_{i,t}$ scores as described in Appendix A.1.

2. Rank the polygons based on these fictitious homicide reductions and classify them into treatment and control groups according to the same criteria used for the actual truce.
3. Estimate the corresponding economic outcomes using these counterfactual treatment and control classifications, pooling the placebo periods alongside the actual truce period within a unified triple-differences estimation.

This procedure allows me to test whether the observed economic improvements could be an artifact of the classification method or driven by underlying endogenous variation in crime and economic activity, rather than by the truce itself. If the methodology were inherently flawed or simply reflecting persistent endogenous relationships, I would observe similar labor market improvements arising from these placebo treatments. Instead, the results show no significant effects in aggregate when the truce is artificially "moved" to pre-truce periods. The lack of any placebo-driven effects strongly suggests that the measured labor market improvements are indeed driven by the actual event of the truce rather than by pre-existing conditions, methodological quirks, or endogeneity in the crime-economic activity relationship.

I provide a full description of the triple-differences estimation strategy in Section 4.1, where the model and identification assumptions are discussed in greater detail.

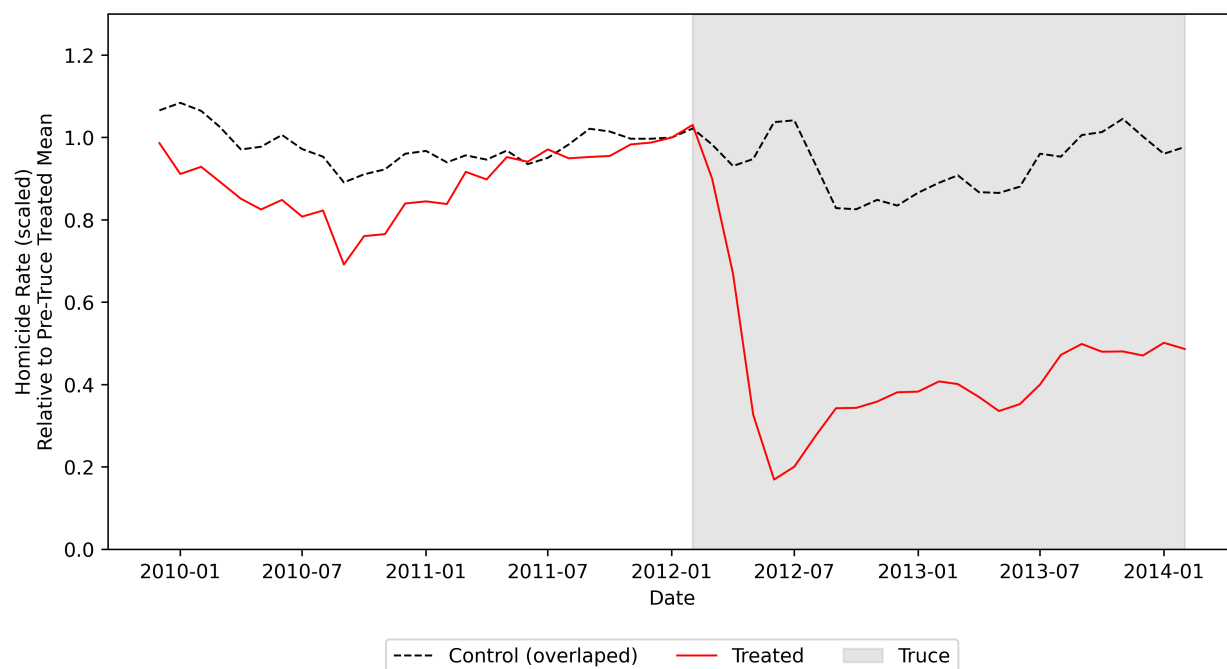
Additionally, I implement an alternative approach to classify treatment areas based on pre-truce gang activity. Instead of defining treatment purely by observed crime reductions during the truce, I use historical data on gang-related homicides to identify areas where gangs were active in the years leading up to the truce. The rationale behind this classification is that these areas were more likely to have been directly affected by the gang negotiations and, consequently, to experience a reduction in violence. By using a measure independent of the actual crime decline during the truce, this approach strengthens the causal interpretation of the results.

This alternative measure, however, is not without limitations. Not all gang-related crimes are perfectly recorded or correctly classified, which introduces noise. Additionally, not all gangs participated in the truce, meaning that some areas continued to experience violence despite the overall decline in homicides. As a result, while this gang-based classification tends to identify polygons that indeed experienced substantial crime declines during the truce, it may fail to capture the full set of impacted areas (see Figure B.12).

3.1.2 Characterizing Treated and Control Areas: Homicide Rates and Economic Conditions

On average, one year after the truce, the homicide rate in treated areas declined markedly. Prior to the truce, these polygons had substantially higher homicide rates than their control counterparts. To better visualize these dynamics, Figure 1 presents normalized homicide rates from January 2010 to January 2014, with both series scaled relative to the average pre-truce rate in treated areas. This allows for a direct visual comparison, with the control group overlapped for reference. Before the truce, the gap between treated and control areas was substantial—about 5.41 homicides per 100,000 inhabitants. Following the onset of the truce, homicide rates in treated areas plummeted by approximately 60%, nearly eliminating the gap. This convergence in violence levels is consistent with a parallel rise in economic activity, as shown in firm entry and employment growth (Figure B.2), suggesting a strong correlation between crime reduction and economic performance.

Figure 1: Normalized Homicide Rates



Note: This figure plots monthly homicide rates per 100,000 inhabitants, normalized to the pre-truce mean in treated areas. The series are smoothed using a centered 3-month moving average. The treated series is shown in red; the control series is overlapped in dashed black. The gray shaded area indicates the duration of the gang truce.

Table 1 provides summary statistics that further distinguish treated and control areas. Consistent with the notion that gangs operate primarily in more densely populated urban and suburban

environments, treated areas feature considerably higher levels of economic activity. For instance, the average number of firms per block in treated areas (22.22) exceeds that of control areas (8.54). Similarly, treated areas host a larger workforce (253 workers per block, compared to 105 in controls) and slightly higher average wages (\$736 versus \$556). These regions also exhibit greater firm turnover, with more frequent openings and closings per quarter.

Table 1: Summary Statistics

Variable	Control		Treatment	
	Mean	SD	Mean	SD
<i>Panel A</i>				
Number of firms per block	8.54	24.67	22.22	82.25
Number of firms per block (excl. top 5%)	8.11	22.87	20.91	76.13
Number of workers per block	104.71	310.31	253.33	750.98
Number of workers per block (excl. top 5%)	55.37	185.2	151.42	627.29
Average wages	556.48	557.16	736.21	551.79
Average wages (excl. top 5%)	500.32	477.5	658.1	447.36
Number of opened firms per quarter	0.26	0.86	0.62	2.66
Number of closed firms per quarter	0.17	0.6	0.43	1.67
Yearly average growth in number of workers (%)	0.51	17.91	1.84	14.65
Yearly average growth in number of workers (%) (excl. top 5%)	2.87	34.97	4.95	37.7
Yearly average growth in number of firms (%)	1.67	28.82	2.26	24.12
Yearly average growth in number of firms (%) (excl. top 5%)	3.08	34.93	5.13	36.21
Yearly average growth of wages (%)	0.27	17.98	1.85	16.67
Yearly average growth of wages (%) (excl. top 5%)	1.66	28.58	2.14	23.42
<i>Panel B</i>				
Homicides Rate (per 100k hab.) before truce	4.22	14.98	9.63	21.51
Homicides Rate (per 100k hab.) after truce	3.93	16.42	4.27	14.89
Number of Blocks (Primary Geographical Unit)	807.0	807.0	774.0	774.0

Notes: Panel A reports the descriptive statistics on business activity, while Panel B provides descriptive statistics on crime, drawing on homicide rates observed 24 months before and after the truce.

4 Identification Strategy

The primary identification strategy employs a standard two-way fixed effects (TWFE) difference-in-differences (DiD) framework. Under this approach, treatment status is defined as a binary indicator $D_i \in \{0, 1\}$, determined by the abnormal reduction in homicides during the truce, where $D_i = 1$ if $Z_{i,T} > M_T$ and 0 otherwise, where $Z_{i,T}$ denotes the measure of homicide reduction for unit i during the truce period (T), and M_T represents the median of $Z_{i,T}$.

To assess the plausibility of the parallel trends assumption required for causal interpretation, I estimate an event-study specification:

$$y_{i,t} = \alpha_i + \gamma_t + \sum_{k=-K}^K \delta_k D_{i,t+k} + \varepsilon_{i,t}, \quad (1)$$

where $y_{i,t}$ is the outcome for unit i at time t , α_i and γ_t are unit and time fixed effects, and $D_{i,t+k}$ are event-time indicators. By examining the coefficients $\hat{\delta}_k$ for periods prior to the treatment, I verify the absence of differential pre-treatment trends.

However, the binary treatment DiD may not fully exploit variation in the intensity of the homicide reduction. To better capture this variation, I extend the analysis by modeling the intensity of the homicide reduction as a continuous dose of treatment, drawing on insights from Callaway, Goodman-Bacon and Sant’Anna (2024). Although the notion of continuous treatment is not new, their framework highlights important sources of bias in TWFE estimation when treatment varies in intensity—particularly when treatment effects are nonlinear.

To operationalize this, I redefine the treatment measure D_i as a continuous variable: let $D_i \in \mathbb{R}_+ \cup \{0\}$, where $D_i = 0$ if $Z_{i,T} \leq M_T$, and $D_i = Z_{i,T} - M_T$ if $Z_{i,T} > M_T$.

To address potential biases from TWFE in this setting, I follow a two-step residualization strategy. First, using only untreated observations—comprising both units that are never treated ($D_i = 0$) and treated units before the truce—I estimate $y_{i,t} = \alpha_i + \gamma_t + \varepsilon_{i,t}$ to obtain $\hat{\alpha}_i$ and $\hat{\gamma}_t$. I then define adjusted outcomes for treated units ($D_i > 0$) as $\Delta \tilde{y}_{i,t} = y_{i,t} - \hat{\alpha}_i - \hat{\gamma}_t$, where the “ Δ ” notation indicates that the outcome has been residualized with respect to unit and time fixed effects, rather than first-differenced over time. This ensures that the empirical object $\Delta \tilde{y}_{i,t}$ corresponds to the population analogue $\Delta Y = Y - \alpha - \gamma$.

Next, I model $\mathbb{E}[\Delta \tilde{y}_{i,t} \mid D_i = d] = \Delta y_t(d)$ using a parametric dose-response function. For instance, a linear form:

$$\Delta y_t(d) = \beta_1 d, \quad (2)$$

or a quadratic form:

$$\Delta y_t(d) = \beta_1 d + \beta_2 d^2 \quad (3)$$

is estimated by OLS.

The marginal effect of treatment intensity is given by the derivative of the dose-response function. In the linear specification, this corresponds to a constant effect $\hat{\beta}_1$, whereas in the quadratic case it varies with intensity as $\hat{\beta}_1 + 2\hat{\beta}_2 d$.

The Average Causal Response (ACR) then averages these marginal effects across treated units:

$$\widehat{ACR} = \frac{1}{N_{D>0}} \sum_{i:D_i>0} \left. \frac{\partial \Delta y_t(d)}{\partial d} \right|_{d=D_i}, \quad (4)$$

which is the empirical counterpart to:

$$ACR = \mathbb{E} \left[\left. \frac{\partial \mathbb{E}[\Delta Y \mid D = d]}{\partial d} \right|_{d=D}, D > 0 \right]. \quad (5)$$

Under the strong parallel trends assumption, the ACR can be interpreted as a causal parameter analogous to a dose-response function. In sum, this continuous-treatment DiD framework allows me to move beyond a binary notion of treatment. While it does not yield a conventional Average Treatment Effect (ATE) or Average Treatment on the Treated (ATT), it provides policy-relevant insights into how incremental reductions in violence translate into improvements in labor market outcomes. This richer characterization hinges on the strong parallel trends assumption, which, although more challenging to verify in a continuous setting, is supported by the fact that when treatment is split into discrete intensity levels, pre-trends are still absent.

4.1 Additional Specifications and Robustness Checks

I implement two alternative identification strategies to assess the robustness of the results (detailed derivations and discussions are presented in Appendix B.1).

First, I consider a gang-related treatment specification where areas are classified as treated if they experienced gang-related homicides before the truce. This alternative definition of treatment poses challenges for standard two-way fixed effects models, as treated and control areas often exhibit dif-

ferential pre-trends—violating the parallel trends assumption. These pre-trends appear to stem from differences in baseline characteristics between treated and control areas—particularly in variables like the number of firms—leading to divergent trajectories prior to the truce. To address this, I follow the method proposed by Borusyak, Jaravel and Spiess (2024), which incorporates group-specific linear time trends based on pre-treatment characteristics. Specifically, I allow the slope of time to vary by baseline firm size. I estimate all fixed effects and trends using only untreated periods and then impute counterfactual outcomes for the treated periods. This approach helps absorb differential trends related to baseline characteristics, providing a cleaner estimate of the treatment effect. A more technical discussion of this procedure and its implications is presented in the appendix B.1.

Second, I implement a triple difference-in-differences (DDD) strategy to test whether the observed effects are specific to the actual truce period or could also emerge spuriously under alternative treatment timings. To do so, I construct placebo datasets in which the truce is hypothetically assigned to begin 3, 5, 7, or 9 months before the real truce’s actual start date. For each placebo timing, I replicate the treatment classification procedure and generate a corresponding dataset. These datasets are then concatenated—along with the actual truce dataset—into a single stacked dataset, with an indicator variable s identifying the scenario: $s = 1$ corresponds to the actual truce, and $s \neq 1$ denotes a placebo.

I then estimate a triple differences model with unit and time fixed effects fully interacted with s . This model isolates the average effect of being classified as treated across all scenarios and estimates the additional effect when treatment status coincides with the real truce period ($s = 1$). Put differently, the identifying variation comes from comparing the effect of treatment in the actual truce period relative to placebo periods. Event study analyses complement this approach by verifying the absence of anticipatory or spurious effects in the placebo periods. The results confirm that significant economic improvements are only observed during the real truce period, strengthening the interpretation that the truce—rather than the classification method—drives the main results.

5 Empirical Findings

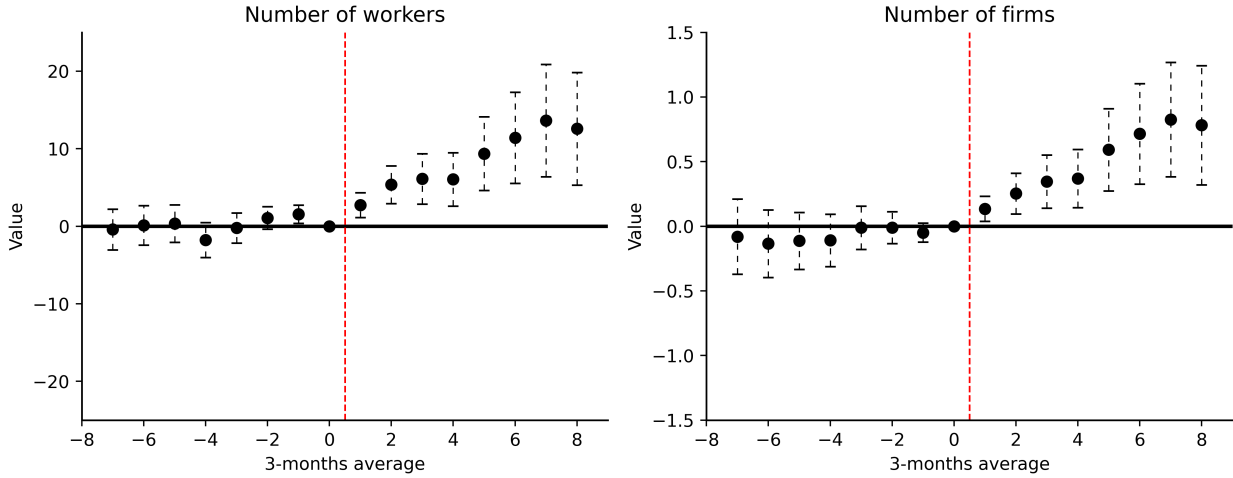
5.1 Baseline Treatment Effects

I begin the empirical analysis by examining how local economic outcomes evolved around the time of the truce. The goal is to assess whether the sharp and exogenous drop in violence coincided with observable changes in employment and firm creation. To this end, I estimate an event study

specification that traces the dynamics of key economic indicators before and after the truce. Figure 2 depicts the truce's effects on both the number of workers and the number of companies in areas that experienced a decrease in crime. The x-axis aggregates data into three-month intervals, covering a span of 24 months both before and after the truce. Within the event study, pre-existing trends in these outputs are indistinguishable, and a significant increase is evident subsequent to the truce between gangs.

The corresponding DiD estimates are presented in Table A.1. For smaller and medium-sized businesses (bottom 95% of firms by size), the truce leads to a statistically significant rise of approximately 8.313 workers per block, representing a 5.49% increase relative to the mean, and an increase of about 0.57 firms per block, translating into a 2.54% uptick. Expressed as elasticities relative to the observed decline in homicides among treated blocks, these estimates imply crime-employment and crime-firm elasticities of approximately 0.11 and 0.05, respectively. By contrast, the top 5% of firms exhibit no statistically significant response; the estimates are close to zero, indicating that these large enterprises do not substantially alter their employment or establishment numbers in response to improved security conditions. This lack of impact for top firms is also evident in an event study analysis that only considers these large companies (Figure A.1), which shows no discernible pre-trends or post-truce changes in this subset. When aggregating all firms together, the overall effect remains positive and significant, with an estimated increase of around 10.9 workers (4.3%) and 0.56 firms (2.4%), suggesting that the bulk of the economic response to reduced crime stems from small and medium-sized enterprises.

Figure 2: Effect of the truce on the number of workers and companies, excluding top 5% largest companies



Notes: This figure displays event-study estimates from Equation (1), focusing on the treatment group selected during the truce period. The X-axis consolidates data into three-month intervals, covering 24 months before and after the onset of the truce. All regressions include unit and time fixed effects, and the sample excludes the top 5% of firms by size. Vertical bars indicate 95% confidence intervals, with standard errors clustered at the geographic block level.

These results are robust to alternative sample splits. In the Online Appendix, I report estimates using different cutoffs for defining treated and control units (e.g., top 40% vs. bottom 40%, top 30% vs. bottom 30%, and so forth). As the sample becomes more selective—focusing on polygons with increasingly extreme crime reductions—the effects grow stronger. For instance, comparing only the top 20% vs. bottom 20% leads to an even larger increase in employment of about 6.4% and a 3.1% rise in the number of firms (see Appendix Table B.2). The corresponding event-study figures confirm no apparent pre-trends, with the divergence emerging only after the truce is implemented (see Appendix Figure B.9).

To further assess the robustness of these findings, I implement sensitivity analyses following the methodology of Rambachan and Roth (2023), which relaxes the parallel trends assumption. Under relative magnitude restrictions, the estimated effects on both workers and firms remain statistically significant even when allowing for post-truce deviations that substantially exceed the magnitude of pre-truce trends. In particular, the breakdown points reach up to four times the maximum pre-truce deviation when evaluating the second-year effect—reflecting the fact that this comparison incorporates two years of pre-truce data. These results indicate that the findings are robust to

sizeable departures from strict parallel trends, especially in the medium run (see Appendix Figures B.7 and B.8).

Under the alternative smoothness restrictions, the estimated effects also remain robust. The results for workers only become insignificant if we allow for deviations from a linear extrapolation of the pre-trend that exceed the equivalent of an entire year of average employment growth in the post-truce period. For firms, the breakdown point corresponds to approximately one-third of a typical year's firm growth. These thresholds suggest that the observed positive effects are not driven by minor trend extrapolation errors and hold even under meaningful departures from the strict parallel trends assumption (see Appendix Figures B.7 and B.8).

These estimates come from formal sector firms, as the data do not capture informal business activity. To provide a broader view, I complement the analysis with national labor market indicators. Household survey data from the World Bank's WDI show that labor force participation rose by about 1.33 percentage points (roughly 2.27%) during the truce, before returning to pre-truce levels once it ended (Figure B.6). This short-lived increase is not mirrored in the regional average for Latin America and the Caribbean, suggesting that the shift in El Salvador reflects a country-specific response likely tied to the temporary decline in violence.

5.2 Robustness Check: Placebo Assignments in Pre-Truce Periods

I redefine the treatment and control groups as if the truce had occurred 3, 5, 7, or 9 months prior to its actual start date, and then reapply the entire selection procedure and estimation strategy to these counterfactual periods. As detailed in the Online Appendix B.1, I implement a triple-differences design that simultaneously considers the actual truce period and these placebo windows. This approach allows me to isolate the portion of the estimated effect attributable to the actual timing of the truce from any pattern that would emerge simply by applying the selection methodology at an arbitrary point in time.

The results, reported in Table B.5, show no significant effects for any of the placebo assignments. The "Treatment (Placebo)" coefficients remain statistically indistinguishable from zero, indicating that no meaningful changes in employment or the number of firms occur when the truce is artificially moved back in time. In contrast, the "Treatment x Truce" interaction consistently yields positive and significant estimates, confirming that the improvements are indeed tied to the actual truce period. The corresponding event-study analysis, illustrated in Figure B.11, similarly reveals no pre-trends or spurious effects under the placebo timings, reinforcing that the observed outcomes emerge only

when the actual truce is in effect.

These placebo results support the conclusion that the main findings are not driven by endogenous selection or methodological artifacts. Instead, the positive labor market responses are associated with the truce’s onset and its subsequent reduction in homicides.

One might expect, based on standard economic theory, that any sharp decline in crime—regardless of timing—would improve local labor market conditions. However, no such pattern emerges in the placebo windows. This apparent discrepancy may reflect the fact that, outside of the truce, sudden and sizable crime drops are rare, and when they do occur, they tend to be noisy or short-lived. As a result, the placebo classifications may fail to capture meaningful shocks large enough to produce detectable economic responses. Supporting this interpretation, Table B.6 presents evidence from a modified approach that restricts the placebo analysis to polygons experiencing large reductions in crime. In these more extreme cases, we do observe a positive association with subsequent employment, consistent with the notion that substantial crime is correlated with economic activity. However, the magnitude of this association remains modest and is substantially smaller than the effects linked to the actual truce period. In Table B.6, the key parameter for causal inference—the “Treatment $\times$ Truce” interaction—remains positive and statistically significant, showing that once the baseline correlation is differenced out, the estimated effect is tied to the actual timing of the truce.

5.3 Robustness Check: Gang-Related Homicide Treatment Classification

To provide further evidence supporting the effects of the truce, I consider an alternative classification: areas are deemed treated if they recorded gang-related homicides before the truce began. This approach aims to capture an exogenous source of treatment intensity, as areas of high pre-truce gang activity experiences larger subsequent declines in violence once the truce takes effect.

Nonetheless, some control areas also experience substantial homicide reductions due to the possibility that crimes were not officially classified as gang-related or the fact that not all gangs participated in the truce. Despite this issue, the results remain robust: the gang-related classification yields significant improvements in both employment and firm counts. Specifically, employment gains are 3.56%, and firm counts increase by 3.03%, as reported in Table B.4. Given the observed reduction in crime within treated polygons, these effects correspond to crime-employment and crime-firm elasticities of approximately 0.34 and 0.29, respectively. While the employment elasticity is not statistically different from previous estimates, the elasticity for firm counts is significantly higher,

aligning with expectations that gang-intensive areas would be particularly responsive to reductions in gang-related violence. This robustness underscores the validity of the main findings and confirms the economic significance of reduced violence.

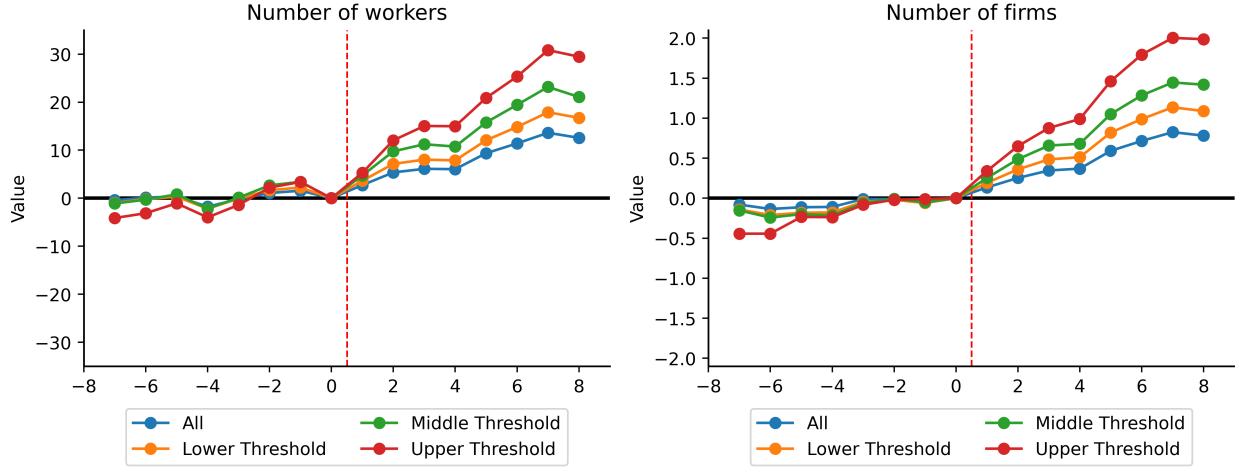
5.4 Continuous Treatment Intensity and Average Causal Response (ACR)

While the binary treatment and control classification provides insights into overall effects, it may obscure the nuanced relationship between the magnitude of homicide reduction and subsequent economic outcomes. To capture this relationship, I treat the reduction in crime as a continuous measure of treatment intensity and estimate the Average Causal Response (ACR). By examining how increments in the “dose” of crime reduction translate into changes in employment and firm counts, this approach can yield a more policy-relevant parameter, especially under the strong parallel trends assumption.

Verifying strong parallel trends is more challenging in a continuous-treatment setting. To address this, I estimate separate event-study regressions using alternative percentile cut-offs of the homicide-reduction distribution among treated polygons. The **High Threshold** corresponds to polygons with homicide reductions at or above the 75th percentile, representing the 25% of areas with the largest crime reductions. The **Intermediate Threshold** includes polygons at or above the 50th percentile, capturing areas in the top half of the distribution. Finally, the **Low Threshold** covers polygons at or above the 25th percentile, which encompasses approximately the top 75% of crime reductions.

These threshold-specific exercises provide suggestive evidence on whether pre-trends remain similar across different intensity ranges. While strong parallel trends cannot be formally verified, the corresponding event-study graphs (see Figure 3) reveal no evidence of differential pre-trends at any of the cut-offs. Furthermore, when modeling the continuous treatment linearly and testing for parallel trends directly (see Appendix Figure A.2), the results confirm that the linear specification satisfies the required assumptions. These findings strengthen the validity of interpreting the ACR estimates as capturing a causal dose-response relationship, driven by the varying intensity of crime reductions.

Figure 3: Effect of the truce by Degrees of Intensity in Declines, excluding top 5% companies



Notes: This figure presents estimated coefficients from Equation (1), applied separately to groups of polygons classified by the intensity of homicide reduction during the truce period. The three lines correspond to Low, Intermediate, and High Threshold groups, defined respectively as polygons at or above the 25th, 50th, and 75th percentiles of the distribution of homicide reductions. The X-axis consolidates time into three-month intervals, spanning 24 months before and after the onset of the truce.

Table 2 reports the estimated ACR for employment and the number of firms, measured one and two years after the truce. Both linear and quadratic specifications produce broadly similar results. A one-standard-deviation reduction in homicide rates—defined relative to each unit’s historical mean and standard deviation of changes—increases the number of workers by about 12.7 to 12.9 (8.4%–8.5%) after one year and roughly 26.1 to 26.5 (17.2%–17.5%) after two years. For the number of firms, the corresponding increases range from approximately 1.0 to 1.03 (4.6%) after one year and about 1.9 to 1.95 (8.7%–8.8%) after two years. These consistent and substantial effects suggest that intensifying the reduction in violent crime delivers meaningful economic gains, with the impact growing more pronounced over time. However, due to the limited duration of the truce, identifying long-term effects in this context remains challenging.

In short, the ACR framework offers a richer perspective on the economic consequences of improved security. Under strong parallel trends, these estimates indicate that even incremental reductions in crime—on the order of one standard deviation—can significantly boost local employment and business activity within one to two years of the truce.

Table 2: Average Causal Response

Dependent	Model	One Year		Two Years	
		Coef/SE	Per. Change	Coef/SE	Per. Change
Workers	Lineal	12.72***	(8.40%)	26.11***	(17.24%)
		(3.71)		(7.16)	
	Quadratic	12.90**	(8.52%)	26.46**	(17.48%)
		(6.49)		(12.50)	
Firms	Lineal	1.01***	(4.55%)	1.93***	(8.68%)
		(0.26)		(0.47)	
	Quadratic	1.03**	(4.61%)	1.95**	(8.78%)
		(0.46)		(0.81)	

Notes: This table reports the Average Causal Response (ACR) estimates as defined in Equation (4), which capture the effect of a one-standard-deviation reduction in crime—measured relative to each unit’s historical mean and standard deviation of changes—on employment (workers) and the number of firms. Results are presented for one and two years after the truce. The models include both linear and quadratic specifications, with coefficients representing the marginal effect of crime reduction. Percentage changes (in parentheses) are calculated relative to the baseline mean. Standard errors are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

6 Mechanisms: Linking Crime Reduction to Labor Demand

To shed light on the mechanisms behind the observed improvements in employment and business outcomes, this section presents evidence in the sequence that most naturally reflects a demand-driven story: beginning with changes in consumer confidence and spending, then tracing firm responses in survival and wage-setting, and finally examining sector-specific effects.

The preceding results suggest that reductions in violent crime are associated with improvements in employment and business. To better understand the forces driving these changes, I outline several potential mechanisms informed by prior literature. These are not exhaustive, but illustrate some of the plausible channels through which crime reduction may spur economic activity. A potential mechanism is that improved security reduces uncertainty for firms, encouraging investment, expansion, and the opening of new businesses—particularly in previously high-risk areas. A second possibility is that safer conditions expand the effective labor supply by enabling workers to commute to or relocate near areas that were previously unsafe, potentially increasing workforce participation and placing downward pressure on wages. A third possibility is that reductions in violence enhance

consumer confidence, encouraging households to spend more freely and engage more actively in the local economy.

These channels are not mutually exclusive, and distinguishing between them is important for informing policy. Understanding whether crime reduction primarily stimulates local consumption, encourages business investment, or facilitates labor mobility can guide more targeted interventions aimed at sustaining economic recovery in high-crime areas. The evidence presented in this study most clearly supports a mechanism in which improved security boosts consumer confidence and local demand, prompting firms to expand operations and increase labor demand. Specifically, I observe (i) stronger consumer confidence—measured using national-level survey data—and higher reported sales among firms in affected areas; (ii) better business conditions, including fewer firm closures; (iii) higher wages, consistent with growing labor demand; and (iv) stronger effects concentrated in non-tradable sectors, which are especially sensitive to local consumer activity. Together, these patterns support the view that crime reduction relaxes demand-side constraints for goods and services in gang-affected communities, allowing local businesses to expand and labor markets to grow.

6.1 Consumer Confidence and Sales Performance

The evidence from the Dynamic Business Survey conducted by FUSADES suggests that sales growth in the designated polygons increased during the truce period, aligning with the reductions in crime observed in my main analysis. Employing my standard event-study specification, I find that the uptick in sales is concentrated only in the truce years. Once the truce ends, sales revert to their pre-truce levels, which is consistent with the notion that benefits may dissipate once violence resumed. The estimated magnitude of the sales increase during the truce corresponds to approximately an 18% improvement in total sales relative to the pre-truce baseline. As shown in the top-left panel of Figure A.3, the event-study estimates highlight this temporal pattern in sales.

While the firm-level survey covers roughly 500 enterprises, they are distributed across only 17 nationally defined polygons, which naturally raises concerns about statistical precision. Nevertheless, the estimates in Table B.7 remain stable under both wild-cluster bootstrap inference and leave-one-out deletions, providing reassurance that the results are not sensitive to small-cluster issues or driven by any single polygon.

Additional supportive evidence, albeit descriptive and more aggregate, emerges from the Economic Outlook Reports of FUSADES (2010–2020). During the truce, the consumer confidence index, produced by the think tank FUSADES, improved at the national level. This index is a composite

measure capturing households' major purchase decisions, and future economic expectations, and the broader economic outlook over both short- and long-term horizons. These patterns are illustrated in the bottom-left panel of Figure A.3. Moreover, when surveyed about the main barriers to investment, firms consistently identified crime as a dominant concern prior to the truce—cited by approximately 40–50% of respondents. During the truce period, this concern fell to around 25%, with "uncertainty" supplanting crime as the primary worry, as depicted in the top-right panel of Figure A.3. After the truce ended, the crime concern indicator rebounded to its pre-truce levels, underscoring the transient nature of the improvement.

Interestingly, not all confidence measures moved in tandem. The business confidence index, which captures firms' current economic perceptions and their expectations, remained largely unchanged during the truce (bottom-right panel of Figure A.3). This discrepancy may reflect the inherent complexity of business sentiment. While firms acknowledged the reduction in crime-related disruptions, other structural uncertainties—about the truce's permanence or broader macroeconomic trajectories—may have tempered any substantial shifts in business confidence.

6.2 Firm Entry and Exit in Safer Areas

A more stable business environment should be reflected in firms' decisions, particularly those involving market exit. One of the clearest signals of improved local conditions is a decline in business closures, as fewer enterprises opt to shut down when consumer confidence rises and profitability prospects brighten. Table A.2 presents results for both firm closures and openings across varying intensity thresholds of crime reduction.

As the treatment intensity increases (from "Treatment" to "Upper Threshold"), the reduction in firm closures becomes more pronounced, reaching over a 13% decrease in the highest-intensity category. These declines in closures represent a substantial portion of the net increase in the number of firms—a key finding, given that new entries alone cannot fully explain the observed expansions in the business base. The data also suggest a modest increase in firm openings, but the dominant driver of growth stems from firms staying in the market longer.

While some of the observed gains could reflect transitions from informality to formality, such effects are likely limited—especially for firm creation—since the main margin of adjustment comes from reduced closures. Moreover, no major enforcement or registration policy changes occurred around 2012 that could account for these patterns.

6.3 Wage Gains Following Crime Reduction

Building on the evidence that firms not only survive longer but also potentially increase their sales under lower-crime conditions, I next examine wage dynamics. In particular, I focus on polygons with at least 10 employees to mitigate noise stemming from very small establishments. This sample refinement allows for a clearer identification of wage increases, which appear most pronounced in areas experiencing substantial economic activity. Such a pattern is consistent with a labor demand-side mechanism: as crime abates and markets become more secure, firms bid up wages to attract and retain labor in tighter local labor markets.

Table A.3 presents the coefficient estimates and percentage changes in wage per capita across different subsets of firms. In the bottom 95% of the firm size distribution, my estimates indicate a statistically significant wage increase of 19.80 dollars, corresponding to a 2.69% increase. Although the coefficient for the top 5% sample is not statistically significant, combining all firms yields a significant overall wage gain of 18.87 dollars, or 2.87%.

These findings align with the notion that the truce boosted local labor demand sufficiently to outweigh any concurrent labor-supply shifts, thereby pushing wages upwards. The event-study patterns (see Figure A.4) corroborate this interpretation: average wages begin to rise after the onset of the truce, and the effect is particularly visible among the bottom 95% of the wage distribution.

In summary, the general evidence supports the idea that a more stable business environment, characterized by falling crime, can tighten local labor markets. As a result, firms appear to raise wages to secure the workforce required to capitalize on improved security conditions and growing demand.

6.4 Sectoral Exposure and Non-Tradable Gains

Beyond the overall increases in employment and firm counts, disaggregating the results by tradable and non-tradable sectors sheds light on the mechanisms driving these expansions. Following the methodology of Knight and Johnson (1997), I classified four-digit CIIU industries as tradable if a substantial portion of their output is sold internationally or if domestic consumption is partly met by imports. I find that the truce did not generate statistically significant effects on tradable sectors in terms of new jobs or new firms (see Figure A.5). In contrast, non-tradable activities—which depend more heavily on local market conditions—show pronounced growth, suggesting that the surge in security primarily boosted local demand. This pattern is especially relevant given that many

smaller enterprises operate in non-tradable sectors, underscoring their vulnerability to high-crime environments and their responsiveness to sudden improvements in security.

While **G** (*Commerce*) and **K** (*Real Estate, Renting, and Business Activities*) contribute the largest absolute increases in employment and firm counts, several traditionally smaller, locally oriented sectors exhibit notably high growth rates. For instance, **F** (*Construction*) experiences a 10.05% rise in employment relative to its pre-truce baseline, while employment in **H** (*Hotels and Restaurants*) and **I** (*Transport and Storage*) grow by 9.85% and 7.73%, respectively (see Figure B.13). These non-tradable sectors are typically indicators of local market vitality, reflecting enhanced consumer confidence, improved conditions for investments, and greater commercial exchange within newly secured areas. Taken together, the evidence reinforces the view that declining violence—in this case, triggered by the temporary truce—can spur immediate, localized gains in economic activity, particularly among smaller firms and service-oriented industries that rely on stable, secure environments.

7 Spillover Effects

A key concern when interpreting the previous results is the potential for *spillovers* between treated and untreated polygons. In principle, spillovers can take both *positive* and *negative* forms. On one hand, improvements in a treated neighborhood—such as reduced crime and heightened economic activity—can *boost* nearby localities via shared consumer demand, commuting flows, or enhanced commercial linkages. On the other hand, economic activity could *shift* away from untreated areas if investors and businesses abandon higher-crime neighborhoods in favor of safer ones, thereby creating negative spillovers.

To initially detect spillovers, I modify the primary difference-in-differences (DiD) specification by adding a dummy variable indicating adjacency to treated polygons. Specifically, I estimate a model of the following form: $Y_{i,t} = \alpha_i + \gamma_t + \tau D_{i,t} + \rho S_{i,t} + \epsilon_{i,t}$, where $Y_{i,t}$ denotes the economic outcome of interest (employment or number of firms) in polygon i at time t , α_i and γ_t are polygon and time fixed effects respectively, $D_{i,t}$ is the treatment indicator, and $S_{i,t}$ captures spillover exposure. I implement two complementary versions of this specification. In the first, $S_{i,t}$ is defined as a dummy equal to one if the polygon is directly adjacent to one treated polygon. In the second, $S_{i,t}$ is defined as a continuous measure of *proximity* to the nearest treated polygon—computed as the negative of the straight-line distance (in kilometers). Hence, higher values of $S_{i,t}$ correspond to closer exposure. Together, these alternative definitions provide complementary evidence on how proximity to treated

areas shapes indirect economic impacts.

This specification allows simultaneous estimation of both direct and indirect effects of the truce: τ captures the direct effect of crime reduction in treated polygons, while ρ identifies spillover effects, which may include both treated and untreated units.

One important caveat is the treatment definition used to identify spillovers. While the main results presented earlier rely on a treatment classification based on observed homicide reductions during the truce, this definition might inadvertently include units already affected by spillovers, potentially misclassifying some adjacent units as directly treated. To mitigate this issue, the spillover analysis in this section employs the alternative treatment definition based solely on the presence of pre-truce gang-related homicides. This provides a more exogenous proxy for treatment. However, even this approach has limitations: some areas treated during the truce may not have previously recorded gang-related homicides, implying that some polygons classified as control may indeed have been treated³. Given this residual uncertainty about whether control units are truly untreated, the estimated spillover effects should be interpreted as an upper bound of their likely magnitude.

Results from this spillover specification (see Table A.4) show that the estimated direct effect of the truce on treated areas remains positive and statistically significant even after accounting for spillover effects. Although the magnitude of the direct effect declines slightly relative to the baseline specification, this difference is not statistically significant. Importantly, the coefficients on the spillover variable are both positive and statistically significant, indicating that polygons adjacent to treated areas also benefited economically from the truce. The estimated spillover corresponds to an increase of approximately 11% in employment and about 6% in the number of firms relative to direct effect of the truce. While these indirect effects are smaller in magnitude than the direct impacts, they nonetheless provide evidence of meaningful economic diffusion beyond the originally treated areas.

8 Aftermath of the Truce: Rising Crime and Economic Reversal

Shortly after the truce dissolved, homicide rates began to rise steadily in both treated and control polygons, eventually surpassing their pre-truce levels in several areas. As shown in Figure B.14, 2015 marked the most violent year in El Salvador's recent history, with the national homicide rate exceeding 100 per 100,000 inhabitants—implying that nearly 0.1% of the population was killed

³Thus, while results from the main definition are also estimated and presented in the Appendix B.8, the analysis here focuses on the more exogenous gang-related homicide classification to interpret spillovers.

that year. This surge in violence coincided with a sharp escalation in the government's anti-gang efforts, often referred to as *mano dura* ("iron-fist") policies, which emphasized aggressive policing. These confrontational tactics intensified conflict between security forces and gang networks, likely fueling further instability and reversing many of the temporary security gains achieved during the truce.

In tandem with rising crime, the economic indicators that had improved during the truce began to reverse. Figure A.6 shows that while the growth in employment levels decelerated, it remained above the pre-truce baseline for most other periods. Similar to the pre-truce period, employment began to follow the same trajectory as its counterfactual, suggesting that once violence resurged, labor market dynamics reverted to their previous patterns. By contrast, the total number of firms steadily declined, ultimately returning to its pre-truce levels by the end of 2015. Interestingly, both treatment and control polygons experienced comparable increases in crime after the ceasefire collapsed, yet the treatment polygons exhibited a stronger negative correlation between rising violence and firm exits. One possible explanation is that these areas had previously benefited more from the truce—experiencing a substantial “peace dividend”—and were therefore more vulnerable to the shock of resurgent insecurity. This suggests that firms in treated areas may have adjusted their operations under the assumption of sustained improvements in security, making them more susceptible to instability when violence returned.

A clear causal link between the end of the truce and the subsequent dip in economic activity cannot be firmly established, in part due to potential confounding from concurrent policy shifts—most notably the escalation of *mano dura* strategies—that may have simultaneously affected crime and economic conditions. Reverse-causality cannot be entirely ruled out, but confounding is the more plausible concern in this context. Nevertheless, two interrelated observations emerge. First, the treatment polygons that experienced considerable improvement during the truce also show a more pronounced deterioration in the number of firms once crime re-escalated, consistent with the hypothesis that smaller or more vulnerable businesses—those which thrived under safer conditions—could not withstand the renewed threats. Second, employment appeared more resilient: although job growth slowed after the truce, it did not fall below pre-truce levels and ultimately returned to its earlier trajectory.

A few years later, another episode of homicide decline occurred in early 2016, following a non-aggression pact among gangs widely understood as a response to the government's intensifying security measures (Gagne, 2017). While the agreement led to a visible drop in the national homicide

rate, crime levels remained close to their pre-truce average, and public sentiment did not shift: consumer confidence and perceptions of safety showed no meaningful gains (see Figure B.16). These two elements—lower violence and improved expectations—are central to the mechanisms proposed in this paper, and their absence may explain why the 2016 episode did not produce comparable economic benefits. Brown et al. (2024), analyzing firm outcomes during that period using data from a large national distributor, finds no positive effects. This is consistent with our evidence showing that large firms and those in the tradable sector were less responsive to reductions in local violence.

9 Discussion

To organize the empirical results, I develop a simple theoretical model with two core ingredients (formal details in Appendix B.3). First, crime raises the effective cost of buying locally in violent areas—a fear-cost borne by consumers—so safer conditions make those purchases relatively cheaper. Second, crime shifts the labor-supply schedule inward—safer conditions shift it outward at a given wage (greater willingness to work holding wages fixed). Local goods are non-tradable, labor can move across locations, and the economic weight of each zone matters for aggregate outcomes. Though deliberately parsimonious, this framework is sufficient to match the main patterns seen during the truce and, more importantly, to speak to two questions: (i) why wages rose rather than fell when crime declined; and (ii) under what conditions employment growth reflects a genuine supply-side expansion rather than a reallocation of jobs across zones.

Prior work (e.g., Utar (2024); Velásquez (2020)) often finds wages rising with violence because higher risk effectively tightens labor supply. The Salvadoran truce looks different: wages rose even as crime fell, and employment expanded. In my framework, this pattern is natural once I account for *two channels*: (i) a **demand** channel—crime raises the effective cost of buying in violent areas; when crime declines, that fear cost drops, local purchases become relatively cheaper, households redirect spending toward those areas, and firms expand hiring; and (ii) a **labor-supply** channel—safer conditions shift the willingness-to-work schedule outward at a given wage. The net effect on wages depends on their relative strength: if the demand expansion is strong enough relative to the outward supply shift, wages rise alongside employment. This aligns with the truce period, where improved security appears to have generated a sizable demand push in formerly violent areas while the supply schedule also shifted outward.

An insight from the model is *scale*: the aggregate impact of improved security is driven by the

economic size of the areas that become safer. When previously violent zones account for a large share of local spending and employment, the relief from fear costs translates into a demand expansion that meets the condition under which wages rise. In El Salvador, the footprint of gang-controlled urban markets appears sizable, making this “size channel” a natural explanation for the wage increases observed during the truce.

A second set of results concerns the composition of employment gains. In my framework, a decline in crime always raises aggregate employment because both forces move in the same direction: the drop in the effective price in formerly violent areas expands local demand, and safer conditions shift the labor-supply schedule outward. Consequently, every crime reduction includes a participation component—part of the aggregate gain comes from new jobs, not just a reshuffling across zones. At the zone level, employment in the formerly high-crime area rises unambiguously, while the response in the low-crime area is ambiguous. When wages increase after crime falls, employment reassigns toward the newly safer area and the low-crime area contracts. This reallocation is stronger when the high-crime market is economically large and consumers are more responsive to safety improvements, and weaker when the labor-supply schedule is highly sensitive to safety (so the supply expansion absorbs part of the demand push).

10 Conclusion

This paper exploits the unique natural experiment provided by the 2012 Salvadoran gang truce to address a critical question: how does reduced violence influence economic growth? The truce, by significantly decreasing homicide rates, provides a rare opportunity to disentangle the causal effects of improved security on local labor markets and business conditions.

The results underscore the transformative potential of crime reduction. A one-standard-deviation drop in homicides leads to an 8.4% increase in employment and a 4.6% rise in the number of firms within the first year. These effects grow over time, with employment and firm counts increasing by 17.2% and 8.7%, respectively, two years after the truce. The mechanisms driving these outcomes appear closely tied to heightened labor demand. Improved consumer confidence, evidenced by increased sales and reduced firm closures, suggests a more favorable business environment during the truce. Although firm openings also rise, the primary driver of net growth is the significant decline in closures. Sectoral analyses further reveal that gains are concentrated in locally dependent industries such as commerce, real estate, and hospitality, highlighting how reductions in crime stimulate

economic activities sensitive to consumer confidence and perceptions of safety.

Interestingly, the analysis also uncovers positive spillover effects. Neighboring areas close to treated polygons exhibit improved employment outcomes, underscoring the broader regional benefits of crime reduction. These findings have important implications for policymakers, as they suggest that targeted interventions in high-crime areas can generate economic benefits that extend beyond their immediate boundaries.

The study also reveals the fragility of these gains. When the truce collapsed, violence resurged, and many of the economic improvements reversed, disproportionately impacting small enterprises. This underscores the critical importance of sustainability in violence-reduction efforts. Without lasting security, the businesses most sensitive to improvements in safety are also the most vulnerable to renewed instability.

In conclusion, this study demonstrates the significant but conditional economic benefits of reducing violence. By providing robust evidence on how improved security fosters local labor demand and business growth, it highlights the critical role of sustainable policies in ensuring long-term economic resilience in high-crime contexts.

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A Appendix

A.1 Construction of the Homicide Reduction Index

To systematically classify treatment and control areas, I construct a standardized measure of abnormal homicide declines. Let $H_{p,t}$ denote the homicide rate in polygon p at month t . I first compute a short-run change in the homicide rate as follows:

$$\Delta H_{p,t} = \left(\frac{1}{36} \sum_{\tau=t-36}^{t-1} H_{p,\tau} \right) - \left(\frac{1}{3} \sum_{\tau=t}^{t+2} H_{p,\tau} \right).$$

This measure $\Delta H_{p,t}$ compares the mean homicide rate over the 36 months prior to t with the mean rate over the subsequent 3 months. During the truce, I expect some polygons to exhibit larger values of $\Delta H_{p,t}$, indicating a sharp reduction in homicide rates.

Next, I standardize each observed drop against the historical distribution of homicide changes within the same polygon. Define the polygon-specific historical mean and standard deviation of these changes (computed over a pre-truce period):

$$\mu_p = \mathbb{E}[\Delta H_{p,t'}], \quad \sigma_p = \sqrt{\mathbb{E}[(\Delta H_{p,t'} - \mu_p)^2]},$$

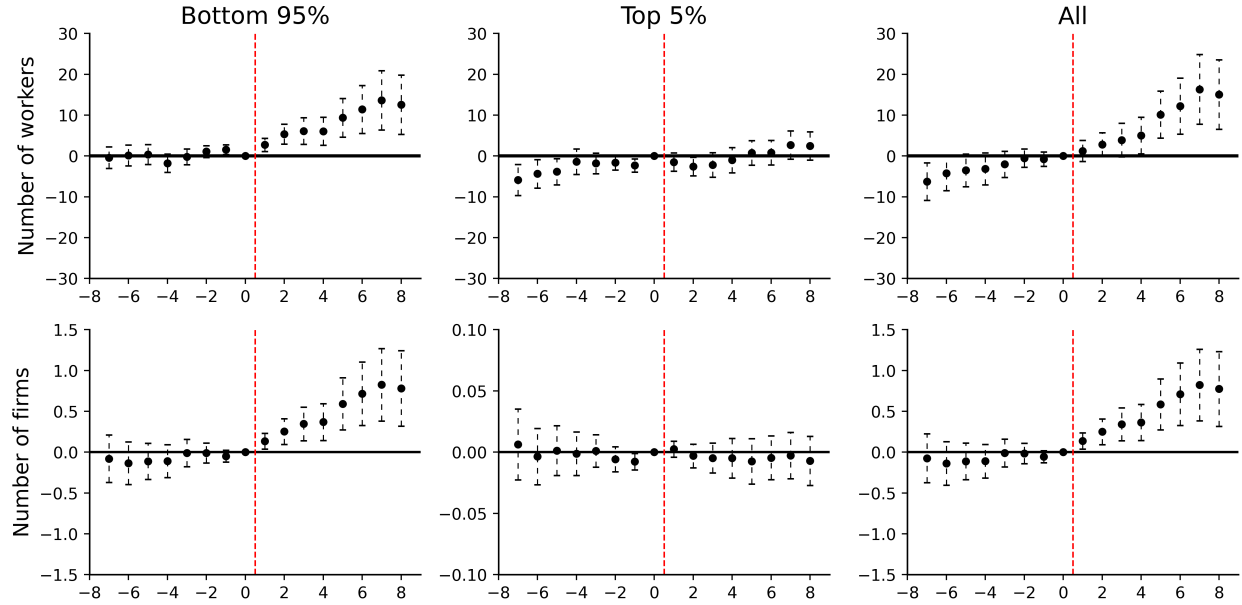
Using these statistics, I construct a standardized measure:

$$Z_{p,t} = \frac{\Delta H_{p,t} - \mu_p}{\sigma_p}.$$

This $Z_{p,t}$ score indicates how unusual the homicide reduction is for polygon p at time t , relative to its own historical variability. A more positive $Z_{p,t}$ means that the current drop in homicides is large relative to what would typically be expected in that polygon based on past fluctuations.

A.2 Figures and Tables

Figure A.1: Event-Study defining Treatment Based on Declines in Homicide Rates During the Truce Period



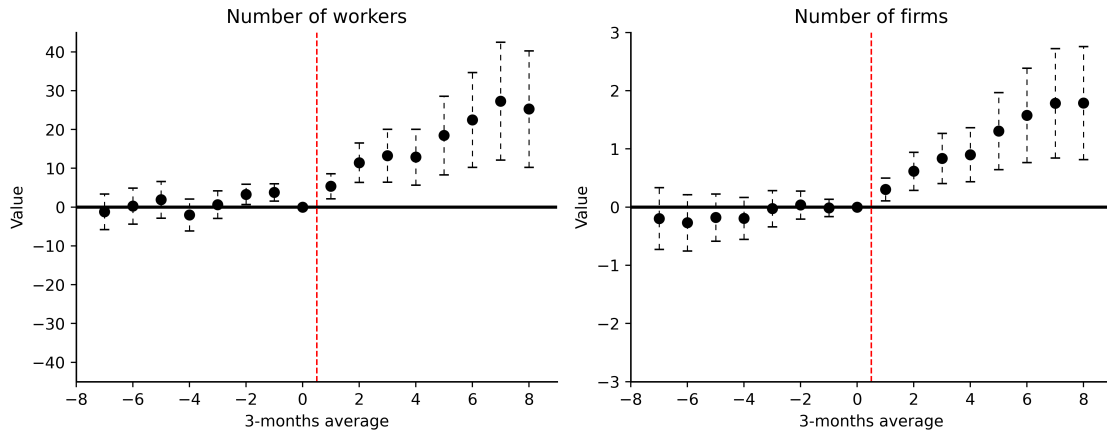
Notes: This figure displays event-study estimates from Equation (1). Treatment is defined as being above the median in the distribution of homicide reductions during the truce period. The X-axis consolidates data into three-month intervals, spanning 24 months before and after the truce. All regressions include unit and time fixed effects. Vertical bars indicate 95% confidence intervals, with standard errors clustered at the geographic block level.

Table A.1: Effect of the truce: Defining Treatment Based on Declines in Homicide Rates During the Truce Period

Variables	Workers		Number of firms	
	Coef/SE	Per. Change	Coef/SE	Per. Change
<i>Small and medium firms: Bottom 95%</i>				
Treatment	8.313*** (2.537)	(5.490%)	0.566*** (0.184)	(2.544%)
R-squared	0.9958		0.9986	
Obs.	25296		25296	
<i>Big firms: Top 5%</i>				
Treatment	2.584 (1.639)	(2.536%)	-0.003 (0.008)	(-0.225%)
R-squared	0.9829		0.9993	
Obs.	25296		25296	
<i>All firms</i>				
Treatment	10.896*** (3.140)	(4.301%)	0.563*** (0.185)	(2.387%)
R-squared	0.995		0.9988	
Obs.	25296		25296	

Notes: This table reports difference-in-differences estimates of the average effect of the truce on employment and the number of firms. Estimates are obtained from a two-way fixed effects (TWFE) specification analogous to Equation (1), using a binary treatment indicator based on whether each unit experienced a homicide reduction above the median during the truce period. Results are shown separately for small and medium firms (bottom 95%) and large firms (top 5%). All regressions include unit and time fixed effects. Standard errors, clustered at the geographic unit, are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A.2: Event-Study Results for Average Causal Response



Notes: This figure presents event-study estimates under a continuous treatment difference-in-differences framework, following the linear specification in Equation (2). Treatment intensity is defined as the standardized reduction in homicides during the truce period. The model includes unit and time fixed effects. Vertical bars denote 95% confidence intervals with standard errors clustered at the geographic block level.

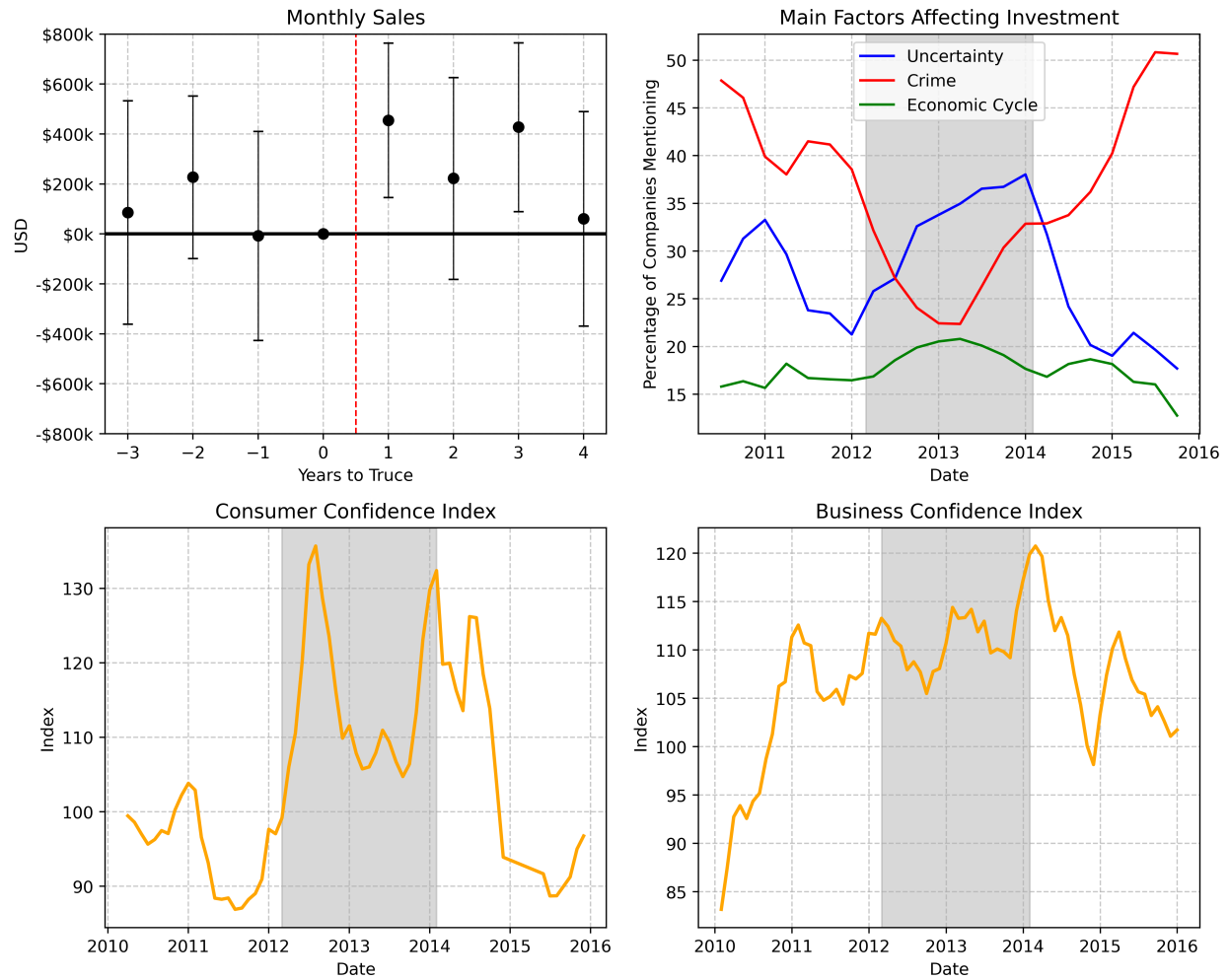
Table A.2: Impact of Crime Reduction on Business Closures and Openings by Treatment Intensity

Variables	Closing business		Opening business	
	Coef/SE	Per. Change	Coef/SE	Per. Change
<i>Treatment</i>				
Treatment	-0.041** (0.018)	(-9.705%)	0.035* (0.019)	(5.739%)
R-squared	0.7825		0.832	
Obs.	25296		25296	
<i>Lower Threshold</i>				
Treatment	-0.059*** (0.022)	(-11.369%)	0.040* (0.023)	(5.421%)
R-squared	0.7905		0.8408	
Obs.	22192		22192	
<i>Middle Threshold</i>				
Treatment	-0.076** (0.030)	(-12.306%)	0.056* (0.030)	(6.393%)
R-squared	0.8056		0.8549	
Obs.	19104		19104	
<i>Upper Threshold</i>				
Treatment	-0.108** (0.051)	(-13.517%)	0.065 (0.048)	(6.672%)
R-squared	0.8324		0.8777	
Obs.	16016		16016	

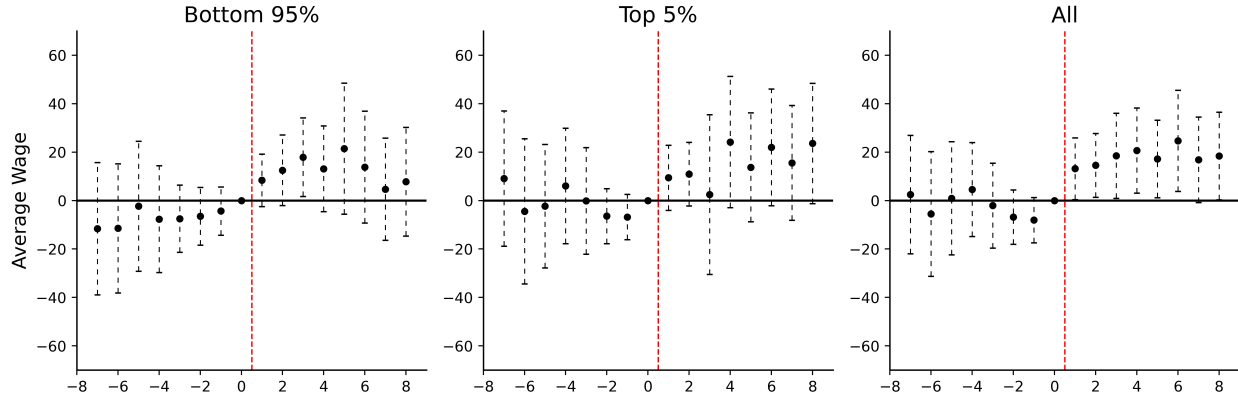
Notes: This table reports difference-in-differences estimates of the average effect of crime reduction on business closures and openings, based on a two-way fixed effects specification analogous to Equation (1). Treatment is defined using thresholds based on the intensity of homicide reduction during the truce period. The “Treatment” row reflects a binary specification comparing treated and control units, while the Lower, Middle, and Upper Thresholds correspond to units at or above the 25th, 50th, and 75th percentiles of homicide reduction, respectively. Each coefficient represents the average effect for units within the corresponding threshold category. All regressions include unit and time fixed effects. Standard errors clustered at the geographic block are shown in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A.3: Sales Growth, Obstacles to Investment, and Confidence Measures During the Truce



Notes: The top-left panel presents event-study estimates of sales growth using an annualized version of Equation (1), where the outcome is average annual sales in polygons affected by the truce. The model includes unit and year fixed effects, and vertical bars denote 95% confidence intervals with standard errors clustered at the polygon level. The top-right panel plots the share of firms citing crime (red), uncertainty (blue), or the economic cycle (green) as the main obstacle to investment, based on firm survey data. The bottom-left and bottom-right panels show consumer and business confidence indices, respectively. Confidence data are sourced from Economic Outlook Reports published by FUSADES (2010–2020); the consumer confidence index is linearly interpolated for the missing period from September 2014 to May 2015.

Figure A.4: Event-Study Results for Wages

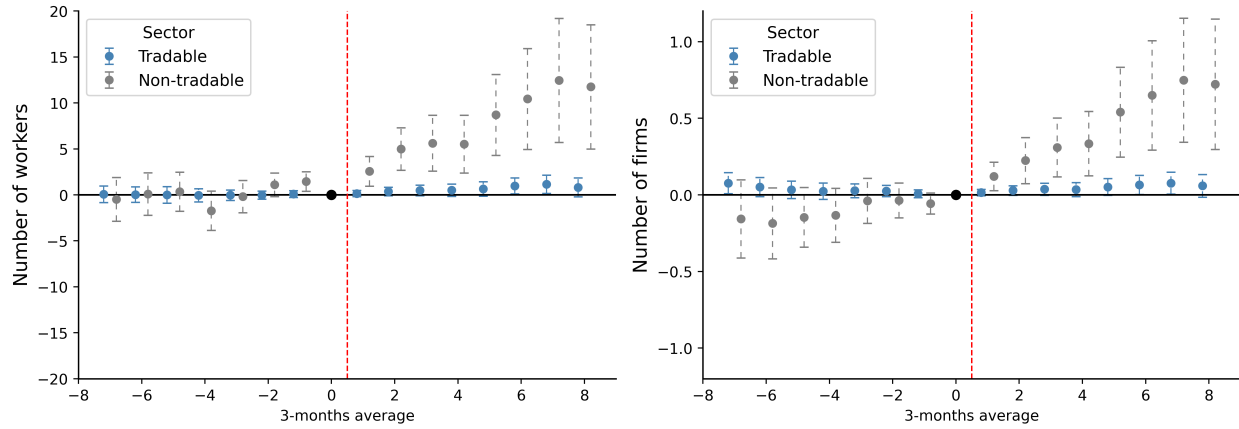
Notes: This figure presents event-study estimates from Equation (1), using wages as the dependent variable. Treatment is defined as being above the median in the distribution of homicide reductions during the truce period. The sample is restricted to polygons with at least 10 employees. Time is grouped into three-month intervals, spanning 24 months before and after the truce. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic block level.

Table A.3: Effect of the truce on Wages: Defining Treatment Based on Declines in Homicide Rates During the Truce Period

Variables	Bottom 95%		Top 5%		All firms	
	Coef/SE	Per. Change	Coef/SE	Per. Change	Coef/SE	Per. Change
<i>Wage per capita</i>						
Treatment	19.813** (8.843)	(2.693%)	15.841 (12.249)	(4.243%)	18.876** (9.393)	(2.871%)
R-squared	0.9547		0.9765		0.8779	
Obs.	11808		11808		11808	

Notes: This table reports difference-in-differences estimates of the average effect of the truce on per-capita wages, using a two-way fixed effects specification analogous to Equation (1). Treatment is defined as experiencing a homicide reduction above the median during the truce period. The sample is restricted to polygons with at least 10 employees. All regressions include unit and time fixed effects. Percentage changes (Per. Change), shown in parentheses, are computed relative to the pre-truce baseline mean. Standard errors are clustered at the geographic unit level and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A.5: Effect of the truce: Tradable vs Non tradable sectors



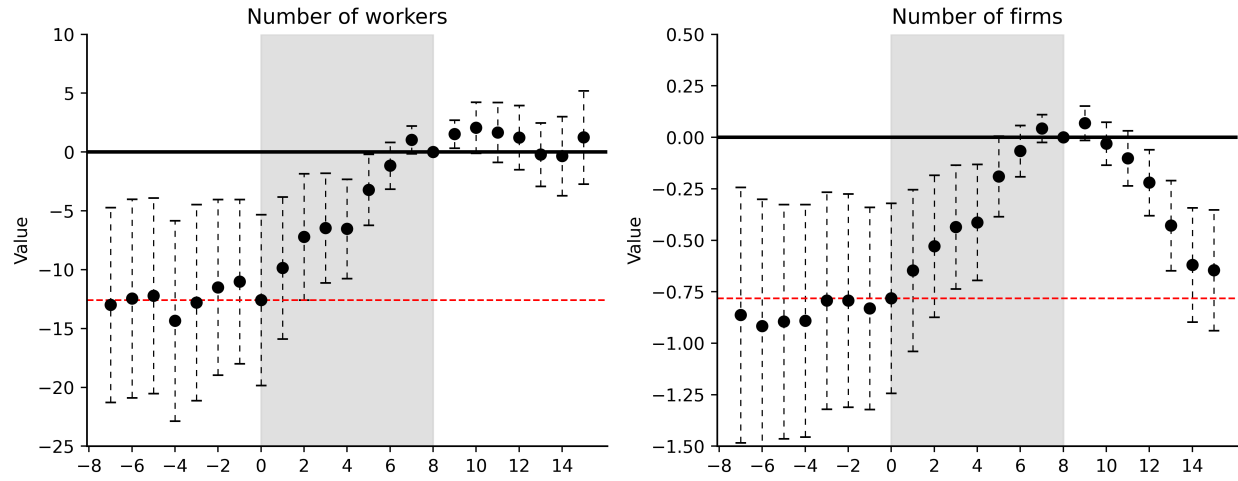
Notes: This figure presents event-study estimates from Equation (1), distinguishing between tradable and non-tradable sectors. Tradable sectors are defined as those more exposed to international trade, while non-tradable sectors primarily serve local markets, following the classification in Knight and Johnson (1997). The model includes unit and time fixed effects, and treatment is defined as experiencing a homicide reduction above the median during the truce period. The X-axis consolidates time into three-month intervals, covering 24 months before and after the truce. Vertical bars indicate 95% confidence intervals, with standard errors clustered at the geographic block level.

Table A.4: Spillover Effects Using Gang-Related Homicide Definition

Variables	Workers	Workers	Workers	Number of firms	Number of firms	Number of firms
	Coef/SE	Coef/SE	Coef/SE	Coef/SE	Coef/SE	Coef/SE
Treatment	6.906* (3.739)	6.394* (3.676)	6.908* (3.741)	0.862*** (0.243)	0.828*** (0.239)	0.862*** (0.243)
Spillover - neighbor		0.730** (0.299)			0.048* (0.029)	
Spillover - proximity (km)			0.176*** (0.054)			0.014*** (0.005)
Obs.	25296	25296	25296	25296	25296	25296

Notes: This table reports spillover effect estimates using an alternative treatment definition based on the presence of gang-related homicides prior to the truce. Estimates are derived from a two-way fixed effects model with unit-specific linear time trends, implemented using the procedure described in Equation (7) in the Appendix. Following Borusyak, Jaravel and Spiess (2024), all fixed effects and trends are estimated using only untreated periods, and treatment effects are computed as the residual between observed and imputed potential outcomes. The variable Spillover – neighbor equals one if a unit shares a geographic border with at least one treated unit. The variable Spillover – proximity is defined as the negative of the straight-line distance (in km) to the nearest treated unit, so that higher values correspond to closer proximity. Standard errors are computed using a block bootstrap procedure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A.6: Trends in the Number of Workers and Firms Post-Truce



Notes: This figure presents event-study estimates from a specification analogous to Equation (1), redefining the event time to center around the end of the gang truce. The sample includes only treated units (above-median homicide reduction during the truce period), and the dependent variables are the number of workers (left panel) and number of firms (right panel). The red dotted line indicates the quarter immediately preceding the truce, which serves as the baseline period. The shaded gray area highlights the duration of the truce. Time is grouped into three-month intervals spanning 24 months after the truce ended. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic block level.

B Online Appendix

B.1 Additional Specifications

B.1.1 Treatment Based on Gang-Related Homicides

To delineate the effects using an alternative treatment definition, I again employed the Two-Way Fixed Effects (TWFE) model, refining the specification by allowing for group-specific time trends based on initial firm size. Specifically, I include an interaction between time and an indicator for polygons whose average number of workers per firm at baseline was above the 60th percentile of the pre-truce distribution. This approach captures potential differences in pre-treatment growth patterns between areas with initially larger and smaller firms. Treatment status D_i is defined by whether a polygon experienced gang-related homicides prior to the truce: $D_i = 1$ if polygon i had gang-related homicides before the truce, and $D_i = 0$ otherwise. The time-varying treatment indicator is then defined as $D_{it} = D_i \times \mathbb{1}\{t \geq t_0\}$, where t_0 denotes the onset of the truce. Hence, $D_{it} = 1$ for treated polygons in all periods at or after the truce, and $D_{it} = 0$ otherwise. The model specification is presented as follows:

$$Y_{it} = \alpha + \tau D_{it} + \gamma_i + \lambda_t + t\omega_g + \epsilon_{it} \quad (6)$$

In this equation, Y_{it} denotes the outcome of interest for each unit i at time t , and D_{it} is the treatment indicator activated post-truce. The terms γ_i and λ_t represent unit and time fixed effects, respectively, while $t\omega_g$ captures a group-specific time trend that allows polygons with initially larger firms (group $g = \text{large}$) to follow a distinct pre-trend relative to smaller ones. The coefficient τ measures the average treatment effect on treated units.

The model was estimated following the imputation-based approach of Borusyak, Jaravel and Spiess (2024). Specifically, I estimate all fixed effects and group-specific trends using only pre-treatment periods (including treated units before treatment onset) and then use these estimates to impute counterfactual outcomes for treated observations in the post-truce period. The estimated treatment effect for each treated observation is then computed as the residual between the observed and predicted outcome:

$$\hat{\tau}_{it} = Y_{it} - \hat{\alpha} - \hat{\gamma}_i - \hat{\lambda}_t - t\hat{\omega}_g \quad (7)$$

Errors are determined using a block bootstrap method.

B.1.2 Placebo tests

The primary modification to the conventional method involves merging multiples data collections, where treatment units are identified based on experiencing an abnormal drop in homicide activity during the truce period, and placebo periods 3,5,7 and 9 months prior to the truce. These datasets are then combined, incorporating a categorical variable s , which indicates whether the assignment to treatment or control occurred during the truce. This variable interacts fully with the unit-fixed effects, time-fixed effects, and the treatment indicator.

The modified TWFE model, incorporating the creation of three datasets and the interaction with a dummy variable for the truce period, is specified as follows:

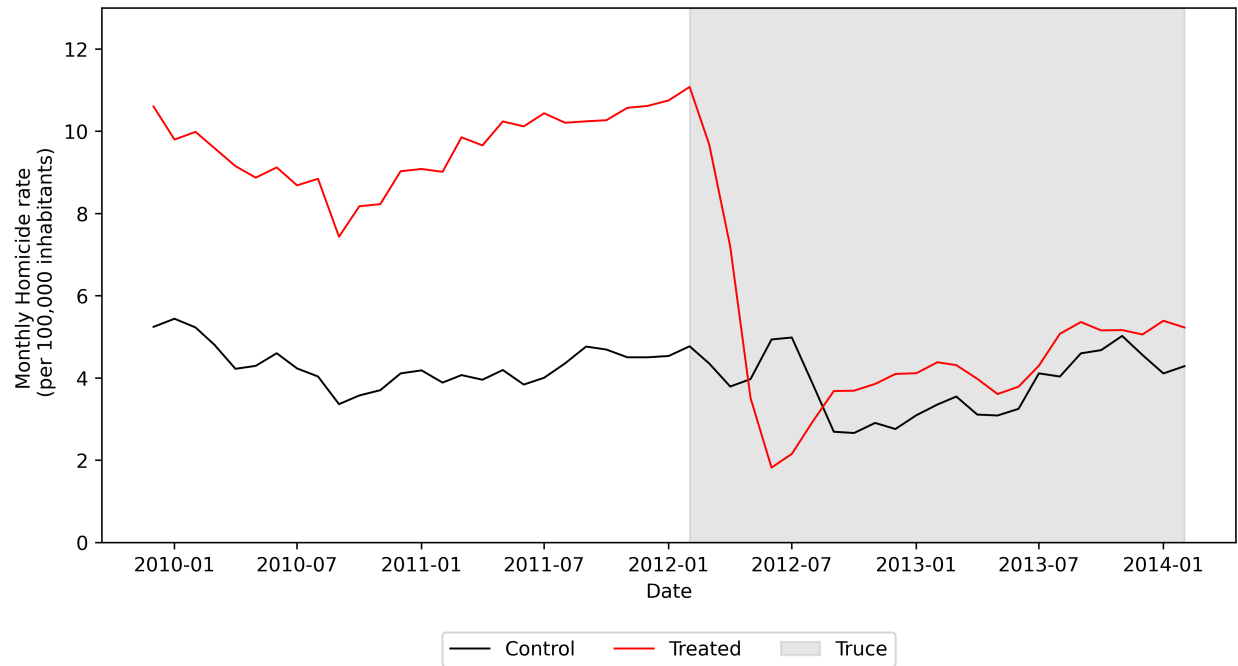
$$Y_{ist} = \alpha + \tau_0 D_{ist} + \tau_1 D_{ist} \mathbb{1}\{s \in Truce\} + \gamma_{is} + \lambda_{ts} + \epsilon_{ist} \quad (8)$$

where Y_{ist} represents the outcome of interest for unit i selected as control or treatment in period s at time t before the event s . D_{ist} is the treatment indicator, activated post-event s for units identified as treated, γ_{is} are the unit fixed effects (polygons) interacted s , λ_{ts} are the time fixed effects interacted s , and ϵ_{ist} is the idiosyncratic error term. The errors are clustered at the level of geographic units interacted with s . The coefficient τ_0 captures the average treatment effect in the placebo scenarios, that is, across periods s preceding the truce when treatment and control units are identified outside the actual intervention window. The coefficient τ_1 measures the incremental effect when the treatment and control selections coincide with the truce period. Hence, the total treatment effect during the real truce is given by the sum $\tau_0 + \tau_1$.

To further dissect the dynamics of the treatment effect, I implement an Event Study analysis. This approach examines the evolution of the impact pre- and post-treatment. The model is augmented with interaction terms between the treatment and time dummies.

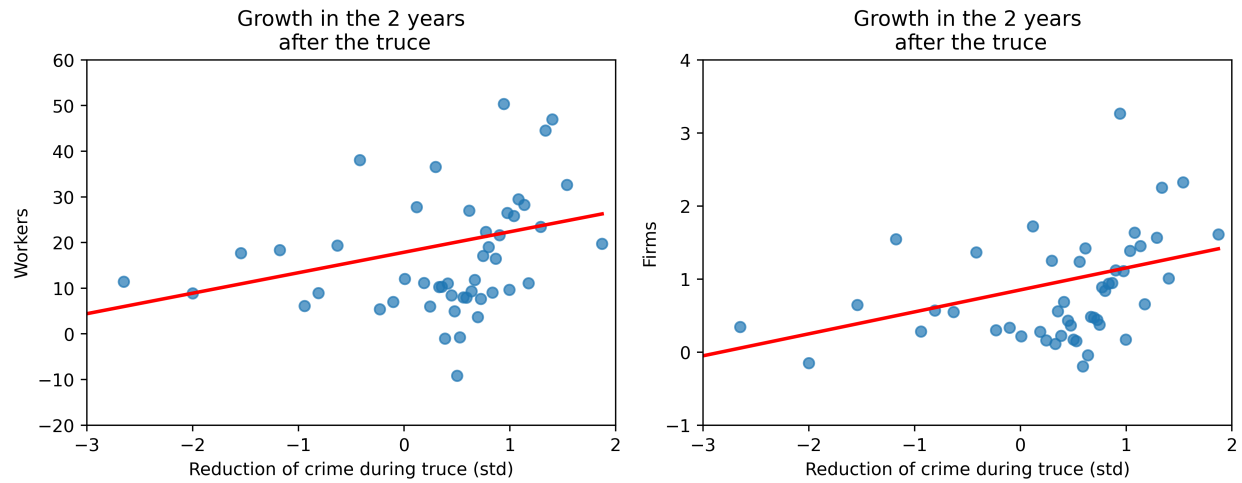
B.2 Figures and Tables

Figure B.1: Homicide Rates in Treated and Control Areas



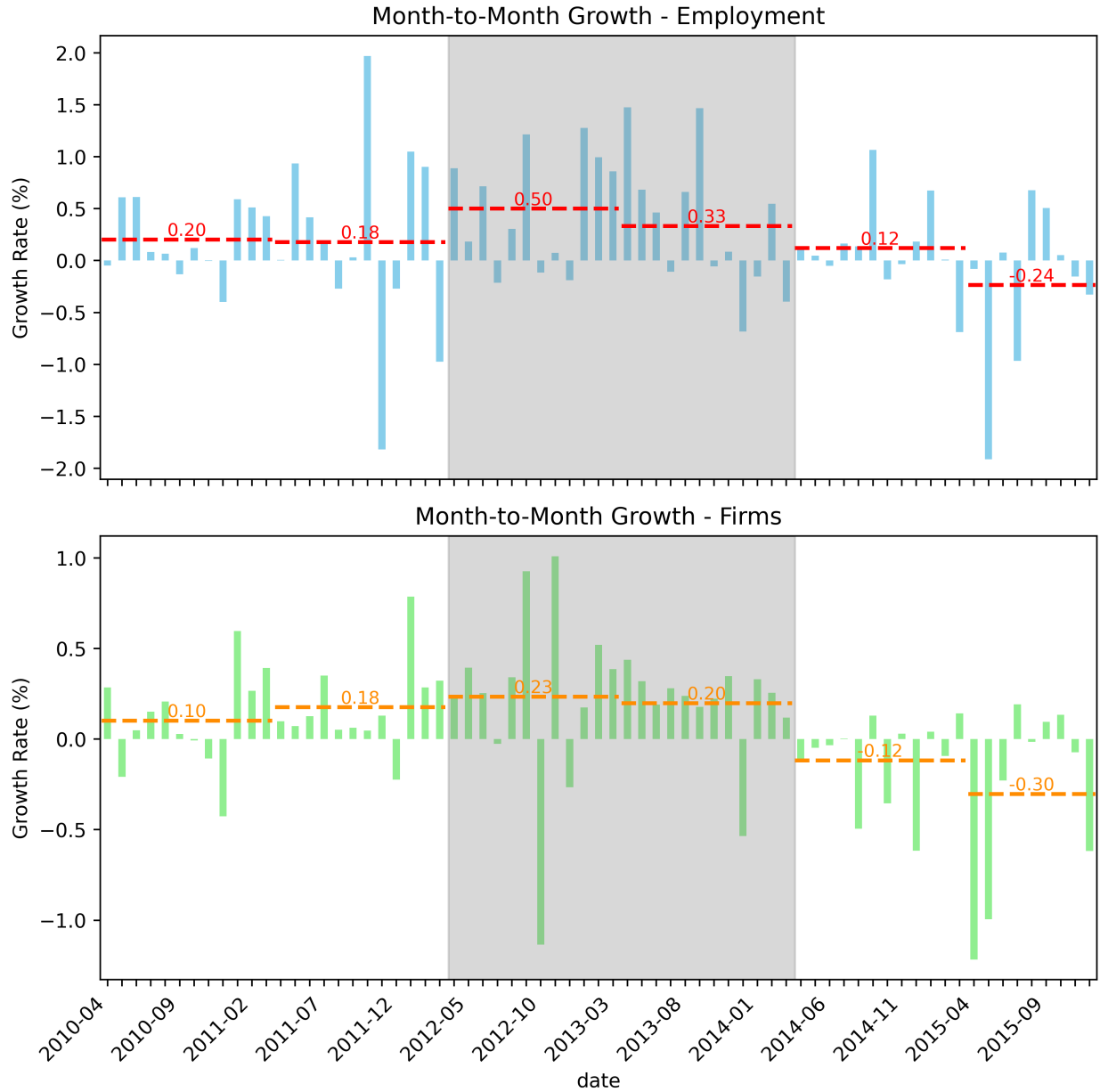
Note: This figure displays the monthly homicide rate per 100,000 inhabitants, computed using a moving average, for areas classified as treated and control. The shaded gray region indicates the truce period.

Figure B.2: Relationship Between Crime Reduction during the truce and Economic activity



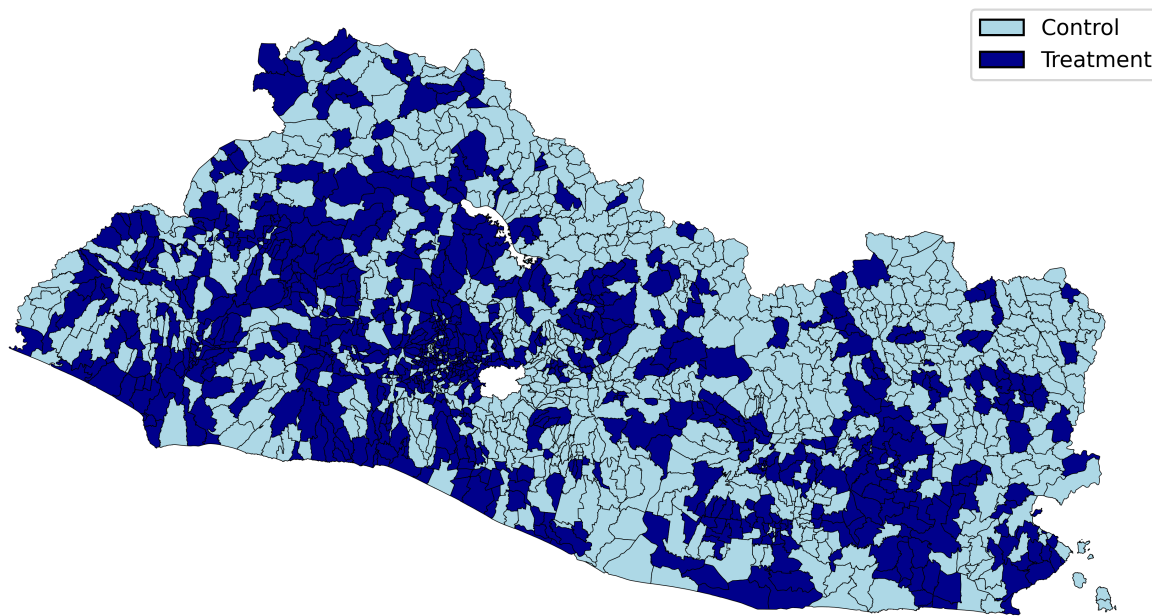
Notes: This figure presents the relationship between the reduction in crime during the truce and two key economic indicators: employment (left panel) and firm presence (right panel). Each point represents a bin of 50 observations, with a fitted linear regression line shown in red. The left panel illustrates the variation in the number of workers, while the right panel depicts the variation in the number of firms, both in response to crime reduction.

Figure B.3: Monthly Growth in Employment and Firms



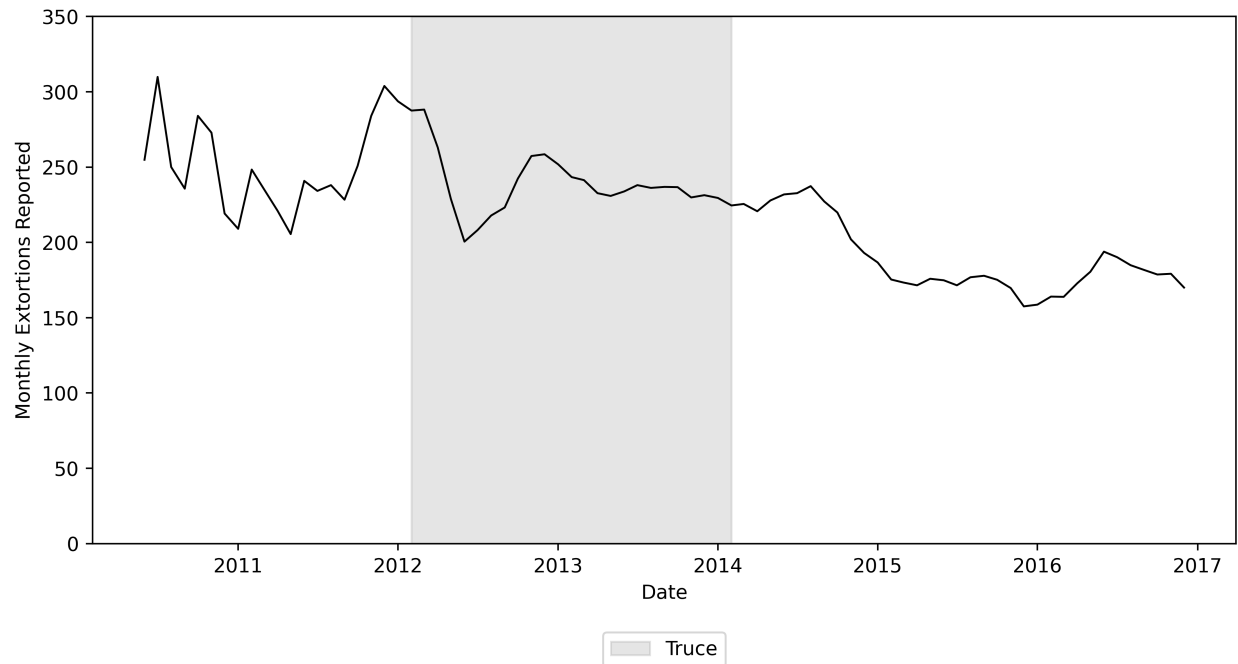
Notes: This figure presents the monthly percentage growth of employment (top panel) and firms (bottom panel) over the study period. Each bar represents the month-to-month growth rate, while the dashed red and orange lines indicate the average growth rates for specific periods. The shaded gray area highlights the truce period.

Figure B.4: Geographic Distribution of Control and Treatment Blocks Based on Crime Reduction During the Truce



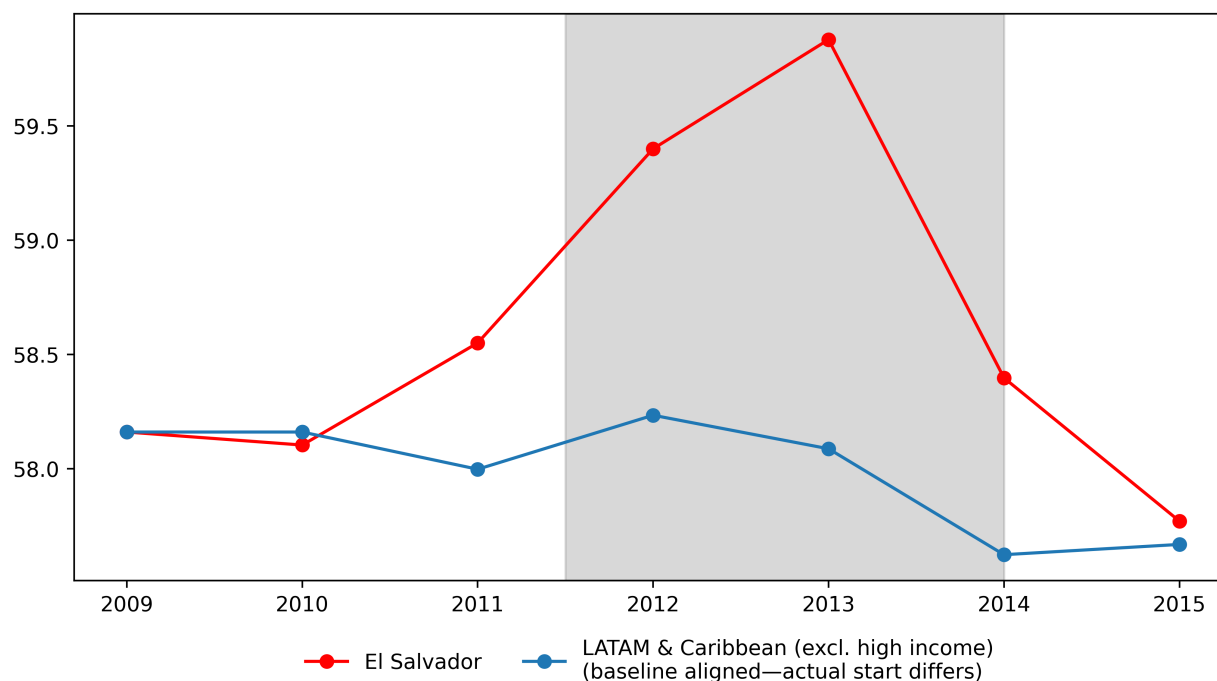
Notes: This map illustrates the spatial distribution of treatment and control blocks across El Salvador, categorized based on the observed reduction in crime rates during the truce period. Areas shaded in blue represent treatment blocks where significant declines in crime were recorded, suggesting these areas were directly influenced by the truce. Conversely, areas in light gray are designated as control blocks, where the truce's impact on crime rates was minimal or non-existent.

Figure B.5: Trends in Monthly Reported Extortions During the Truce Period



Notes: This graph illustrates the trend in monthly reported extortions from 2011 to 2017, with a specific focus on the truce period highlighted in grey. Despite the implementation of the truce, aimed at reducing gang violence, the data indicates that extortion rates remained relatively stable throughout the truce period.

Figure B.6: Employment to Population Ratio (%) Over Time using WDI data



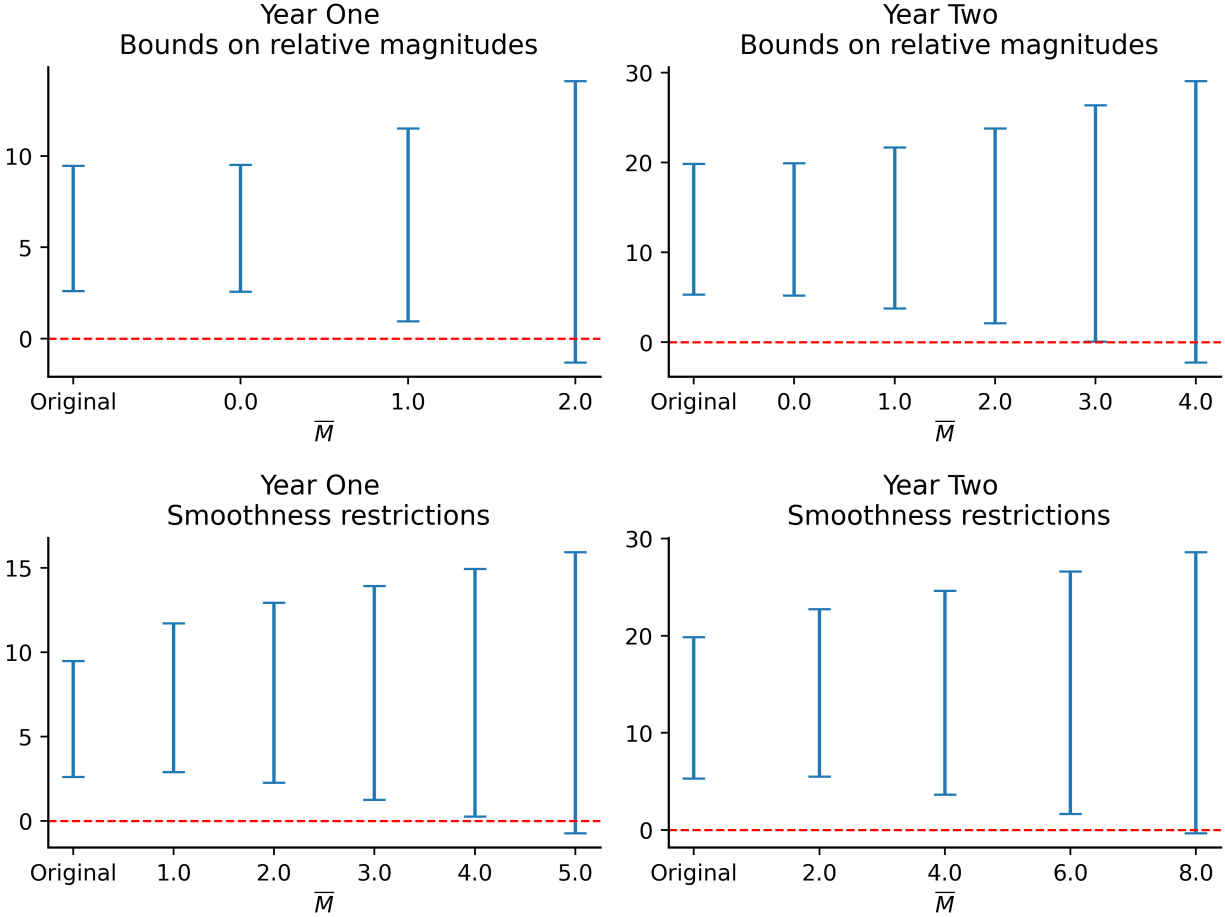
Notes: This figure presents a key labor market indicator for the period 2009–2015, using data from the World Development Indicators (WDI), which are based on household surveys. The graph displays the employment-to-population ratio (15+, total %), illustrating the proportion of the working-age population engaged in employment each year. The shaded area corresponds to the period of the gang truce (2012–2014). For comparison, the figure also includes the regional trend for Latin America and the Caribbean (excluding high-income countries), baseline-aligned to El Salvador’s truce period.

Table B.1: Employment to Population Ratio Over Time using WDI data

Region	Year-by-Year							Summary		
	2009	2010	2011	2012	2013	2014	2015	Pre-Truce	Truce	Difference
El Salvador	58.16	58.10	58.55	59.40	59.88	58.40	57.77	58.27	59.22	0.95
LATAM and Caribbean (excl. high income)	60.16	60.16	60.00	60.24	60.09	59.63	59.67	60.11	59.99	-0.12
Difference	-2.00	-2.06	-1.45	-0.84	-0.21	-1.23	-1.90	-1.84	-0.76	1.08

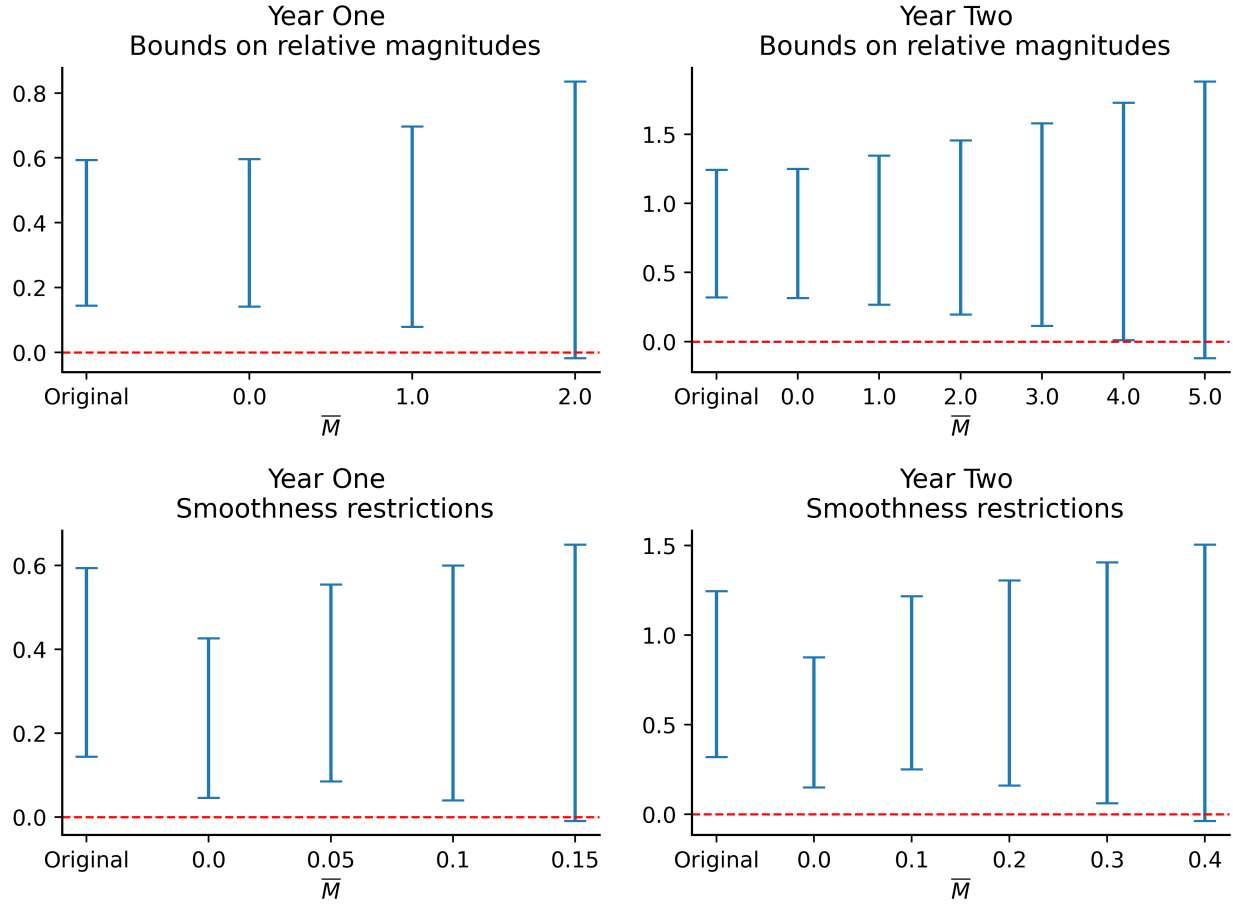
Notes: This table presents employment rate for the period 2009–2015, using data from the World Development Indicators (WDI), which are based on household surveys. The employment-to-population ratio (15+, total %), is the proportion of the working-age population engaged in employment each year. The truce period spans 2012–2014, and the pre-truce period corresponds to the earlier years shown in the table (2008–2011).

Figure B.7: Robust Confidence Intervals for the Effect on the Number of Workers: Sensitivity to Violations of Parallel Trends



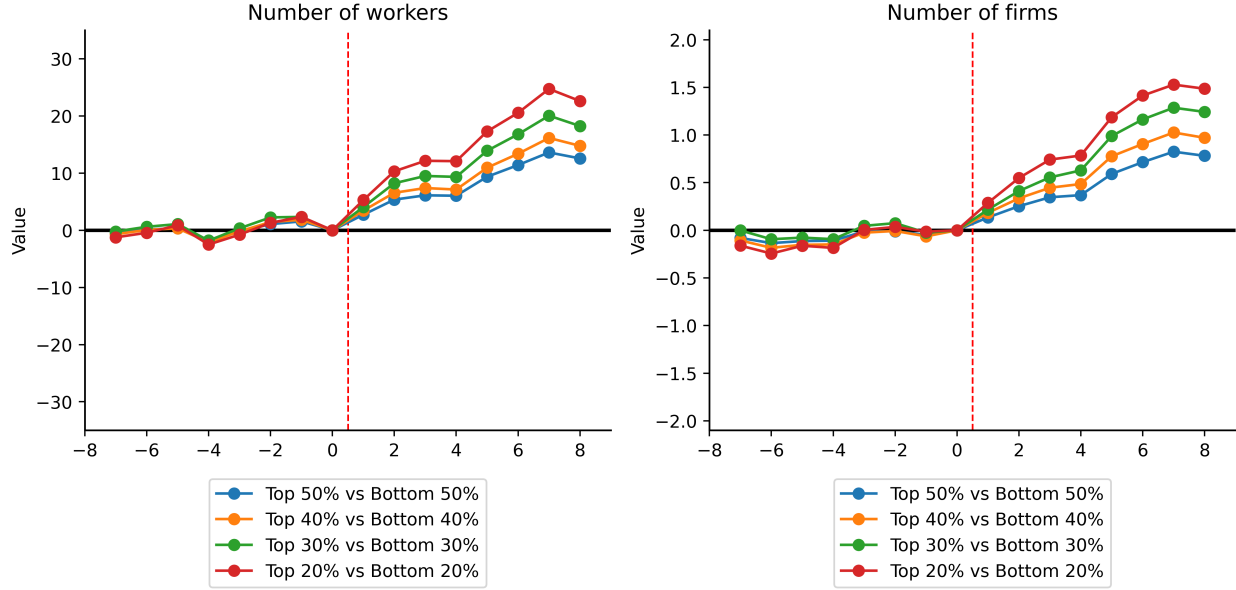
Notes: This figure reports robust 95% confidence intervals for the estimated treatment effects at year 1 and year 2 post-treatment using the methodology developed by Rambachan and Roth (2023). The top panels present results under **relative magnitude restrictions**, where $\bar{M}$ denotes the maximum size of post-treatment deviations from parallel trends relative to the worst violation observed pre-treatment. For example, $\bar{M} = 2$ allows post-treatment violations up to twice as large as the maximum observed pre-treatment deviation. The bottom panels present results under **smoothness restrictions**, where $\bar{M}$ represents the maximum allowed change in the slope of the trend across consecutive post-treatment periods. In all panels, the label "Original" corresponds to the confidence interval under the exact parallel trends assumption.

Figure B.8: Robust Confidence Intervals for the Effect on the Number of Firms: Sensitivity to Violations of Parallel Trends



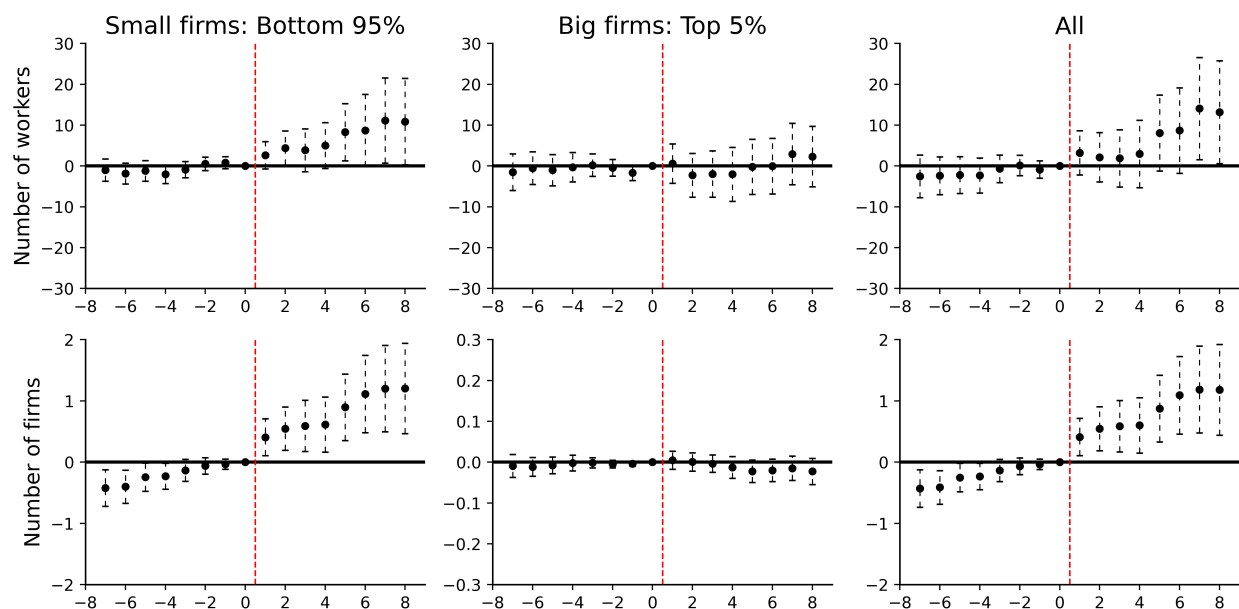
Notes: This figure reports robust 95% confidence intervals for the estimated treatment effects at year 1 and year 2 post-treatment using the methodology developed by Rambachan and Roth (2023). The top panels present results under **relative magnitude restrictions**, where $\bar{M}$ denotes the maximum size of post-treatment deviations from parallel trends relative to the worst violation observed pre-treatment. For example, $\bar{M} = 2$ allows post-treatment violations up to twice as large as the maximum observed pre-treatment deviation. The bottom panels present results under **smoothness restrictions**, where $\bar{M}$ represents the maximum allowed change in the slope of the trend across consecutive post-treatment periods. In all panels, the label "Original" corresponds to the confidence interval under the exact parallel trends assumption.

Figure B.9: Event-Study Results for Different Treatment-Intensity Thresholds



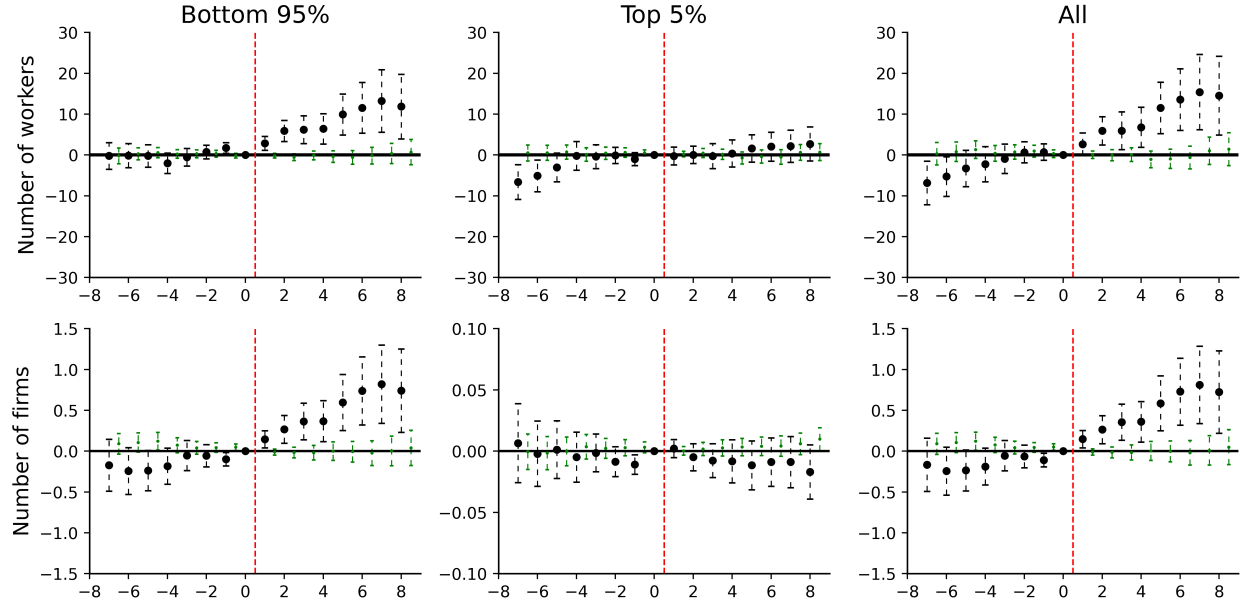
Notes: This figure presents event-study estimates based on Equation (1), comparing outcomes across treated units with varying degrees of treatment intensity. Specifically, it plots separate estimates for units in the top 50% vs. bottom 50%, top 40% vs. bottom 40%, and top 20% vs. bottom 20% of the homicide reduction distribution during the truce period. The X-axis aggregates data into three-month intervals spanning 24 months before and after the truce. All regressions include unit and time fixed effects.

Figure B.10: Event-Study defining Treatment Based on Gang-Related Homicides Before the Truce



Notes: This figure presents event-study estimates using an alternative treatment definition based on the presence of gang-related homicides prior to the truce. The analysis follows the methodology in Borusyak, Jaravel and Spiess (2024), as formalized in Equation (7). Time is grouped into three-month intervals, spanning 24 months before and after. Vertical bars represent 95% confidence intervals, with standard errors calculated using a block bootstrap procedure for post-truce periods and OLS with clustering at the unit level for pre-truce periods.

Figure B.11: Event-Study Results for Placebo Truce Assignments



Notes: This figure presents event-study estimates based on the specification in Equation (8), which compares the estimated treatment effects for the actual truce period versus placebo periods starting 3, 5, 7, and 9 months prior. Each dataset defines treatment based on abnormal homicide reductions, and is stacked with a scenario indicator s identifying the truce or placebo period. Unit and time fixed effects are fully interacted with s . The X-axis consolidates time into three-month intervals spanning 24 months before and after the treatment assignment. Black lines represent the estimated effects for the actual truce, while green lines display the average placebo effects across all alternative scenarios. Vertical bars represent 95% confidence intervals, with standard errors clustered at the unit $\times$ scenario level.

Table B.2: Sensitivity of Treatment Effects to Alternative Sample Cutoffs

Variables	Workers		Number of firms	
	Coef/SE	Per. Change	Coef/SE	Per. Change
<i>Treatment</i>				
Treatment	8.313*** (2.537)	(5.490%)	0.566*** (0.184)	(2.544%)
R-squared	0.9958		0.9986	
Obs.	25296		25296	
<i>Top 40% vs Bottom 40%</i>				
Treatment	9.884*** (3.130)	(5.666%)	0.727*** (0.225)	(2.854%)
R-squared	0.9959		0.9987	
Obs.	19808		19808	
<i>Top 30% vs Bottom 30%</i>				
Treatment	11.939*** (4.053)	(5.778%)	0.833*** (0.289)	(2.785%)
R-squared	0.9961		0.9987	
Obs.	14848		14848	
<i>Top 20% vs Bottom 20%</i>				
Treatment	15.681*** (5.814)	(6.380%)	1.088*** (0.395)	(3.083%)
R-squared	0.9962		0.9989	
Obs.	9920		9920	
<i>Top 10% vs Bottom 10%</i>				
Treatment	9.997** (4.297)	(5.898%)	0.677 (0.415)	(2.564%)
R-squared	0.993		0.9965	
Obs.	5104		5104	

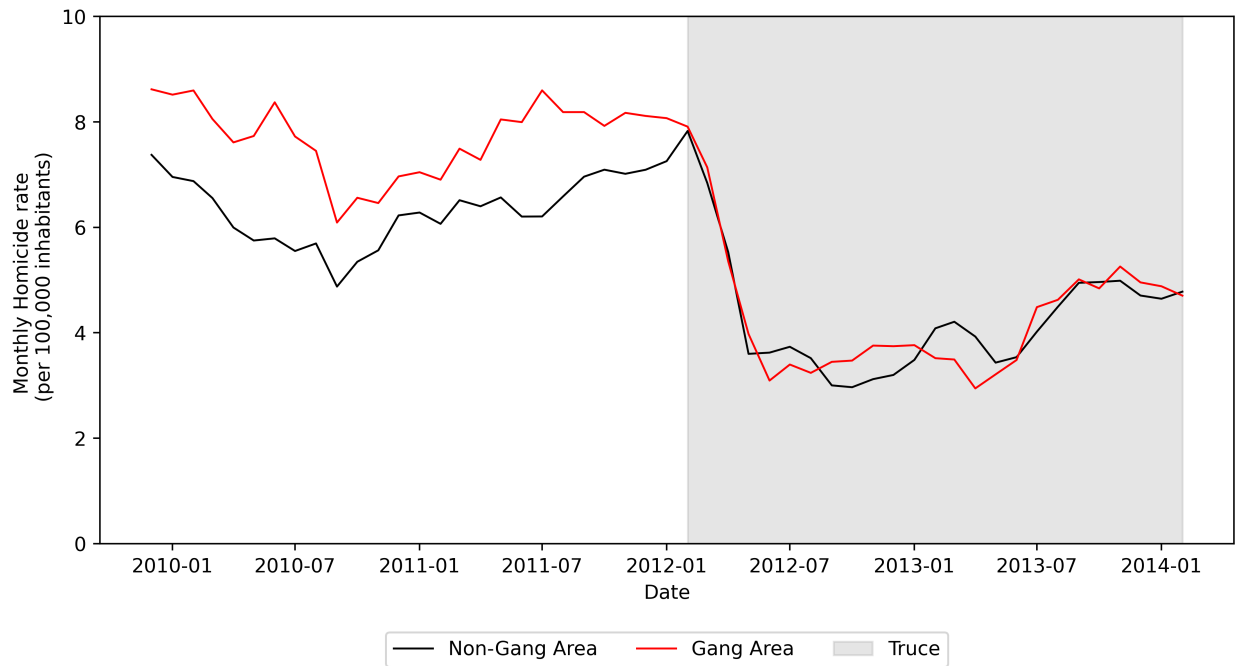
Notes: This table reports difference-in-differences estimates for the effect of the truce on the number of workers and firms, using a two-way fixed effects specification analogous to Equation (1). Each column corresponds to a different treatment cutoff, comparing units in the top versus bottom $x\%$ of homicide reductions during the truce period (e.g., 50% vs 50%, 40% vs 40%, etc.). The sample excludes the top 5% of firms by size. All regressions include unit and time fixed effects. Percentage changes (Per. Change), shown in parentheses, are computed relative to the pre-truce baseline mean. Standard errors are clustered at the level of geographic units. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.3: Summary Statistics: Defining Treatment Based on Gang-Related Homicides Before the Truce

Variable	Area without gangs		Area with gangs	
	Mean	SD	Mean	SD
<i>Panel A</i>				
Number of firms per block	6.55	23.12	28.42	90.23
Number of firms per block (excl. top 5%)	6.21	21.1	26.77	83.61
Number of workers per block	79.28	272.91	326.47	826.45
Number of workers per block (excl. top 5%)	42.18	172.02	193.76	690.09
Average wages	514.7	541.89	841.38	533.38
Average wages (excl. top 5%)	471.82	481.49	738.04	400.92
Number of opened firms per quarter	0.19	0.83	0.81	2.92
Number of closed firms per quarter	0.11	0.53	0.57	1.85
<i>Panel B</i>				
Homicides Rate (per 100k hab.) before truce	6.38	20.77	7.61	14.87
Homicides Rate (per 100k hab.) after truce	4.1	17.98	4.09	11.35
Number of Blocks (Primary Geographical Unit)	953.0	953.0	628.0	628.0

Notes: Panel A reports the descriptive statistics of the business base and panel B reports the descriptive statistics of the crime base using homicide rates 24 months before and after the truce into consideration.

Figure B.12: Homicides rates in Treated Areas and Control Areas: Defining Treatment Based on Gang-Related Homicides Before the Truce



Note: This graph illustrates the monthly homicide rate per 100,000 inhabitants, using a moving average. Additionally, the graph features a shaded gray area to delineate the period of the truce between gangs.

Table B.4: Effect of the truce: Defining Treatment Based on Gang-Related Homicides Before the Truce

Variables	Workers		Number of firms	
	Coef/SE	Per. Change	Coef/SE	Per. Change
<i>Small and medium firms: Bottom 95%</i>				
Treatment	6.90* (3.75)	(3.56%)	0.86*** (0.24)	(3.03%)
Obs.	25296		25296	
<i>Big firms: Top 5%</i>				
Treatment	-0.06 (2.64)	(-0.04%)	-0.01 (0.01)	(-0.58%)
Obs.	25296		25296	
<i>All firms</i>				
Treatment	6.85 (4.62)	(2.10%)	0.85*** (0.25)	(2.83%)
Obs.	25296		25296	

Notes: This table reports difference-in-differences estimates using an alternative treatment definition based on the presence of gang-related homicides prior to the truce. The estimation follows the methodology of Borusyak, Jaravel and Spiess (2024), as formalized in Equation (7). The sample includes both small and medium firms (bottom 95%) and large firms (top 5%). All regressions include time and units fixed effects, and standard errors are computed using a block bootstrap procedure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.5: Effect of the truce: Placebo Triple-Differences Estimates

Variables	3 months be- fore	5 months be- fore	7 months be- fore	9 months be- fore	All placebos
<i>Workers</i>					
Treatment (Placebo)	-0.387 (2.060)	-0.112 (1.841)	0.038 (1.610)	-0.641 (1.394)	-0.275 (0.872)
Treatment x Truce	8.700*** (3.267)	8.424*** (3.134)	8.274*** (3.004)	8.953*** (2.894)	8.588*** (2.682)
R-squared	0.9961	0.9964	0.9965	0.9966	0.9968
Obs.	50592	50592	50592	50592	126480
<i>Number of firms</i>					
Treatment (Placebo)	0.102 (0.170)	-0.146 (0.160)	-0.129 (0.155)	-0.108 (0.148)	-0.070 (0.079)
Treatment x Truce	0.463* (0.250)	0.711*** (0.244)	0.694*** (0.241)	0.674*** (0.236)	0.636*** (0.200)
R-squared	0.9987	0.9987	0.9988	0.9988	0.9988
Obs.	50592	50592	50592	50592	126480

Notes: This table reports the triple-differences estimates examining the effect of the truce and several placebo periods set 3, 5, 7, and 9 months prior to the actual onset of the truce. Each column presents results from a separate specification. The rows labeled “Treatment (Placebo)” capture the hypothetical impact of assigning treatment status using the same procedure but well before the truce actually occurred. The row “Treatment x Truce” represents the additional effect observed exclusively during the actual truce period. Standard errors (in parentheses) are clustered at the polygon level interacted with placebo period or truce. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table B.6: Effect of the truce: Placebo Triple-Differences Estimates by Treatment Intensity

Variables	Treatment	Lower Threshold	Middle Threshold	Upper Threshold
<i>Workers</i>				
Treatment (Placebo)	-0.275 (0.872)	0.363 (0.952)	1.830 (1.142)	3.638*** (1.389)
Treatment x Truce	8.588*** (2.682)	10.573*** (3.386)	12.239** (4.787)	16.613* (8.862)
R-squared	0.9968	0.9968	0.997	0.9968
Obs.	126480	111024	95616	80256
<i>Number of firms</i>				
Treatment (Placebo)	-0.070 (0.079)	-0.011 (0.085)	0.104 (0.101)	0.164 (0.124)
Treatment x Truce	0.636*** (0.200)	0.814*** (0.244)	0.921*** (0.333)	1.283** (0.559)
R-squared	0.9988	0.9989	0.9989	0.9989
Obs.	126480	111024	95616	80256

Notes: This table reports the triple-differences estimates examining the effect of the truce and several placebo periods set 3, 5, 7, and 9 months prior to the actual onset of the truce. Each column presents results from a separate specification. The “Treatment” column presents the baseline results, while the lower, middle, and upper thresholds apply the same treatment-intensity cutoffs used in the main text—namely, the 25th, 50th, and 75th percentiles of the homicide-reduction distribution, respectively. The rows labeled “Treatment (Placebo)” capture the hypothetical impact of assigning treatment status using the same procedure but well before the truce actually occurred. The row “Treatment x Truce” represents the additional effect observed exclusively during the actual truce period. Standard errors (in parentheses) are clustered at the polygon level interacted with placebo period or truce.

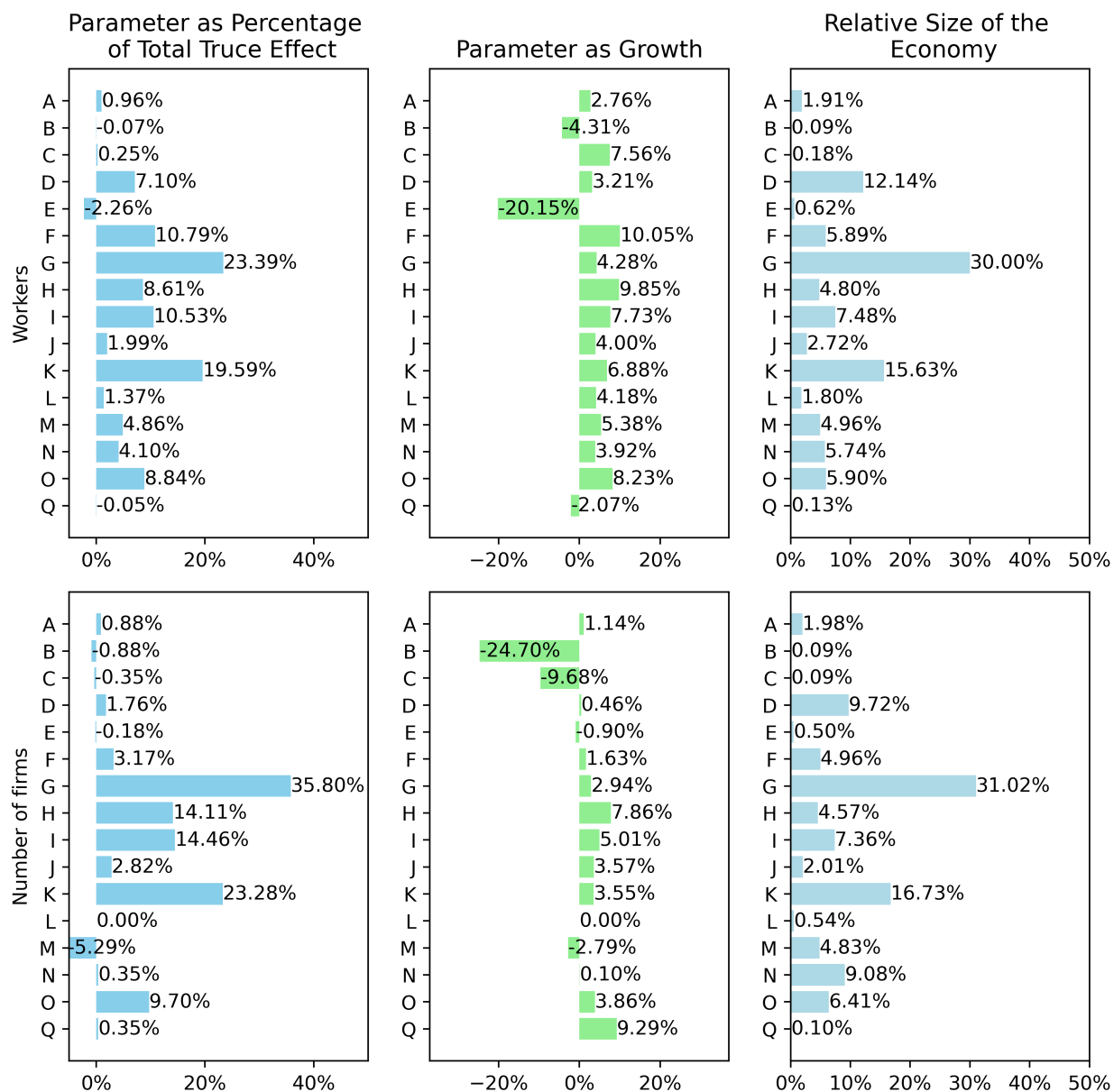
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table B.7: Monthly Sales Results and Robustness to Excluding Individual Polygons

$\hat{\beta}$	Classical p	Wild p	Firms	Obs.	Clusters
Full Sample					
291862.1	0.0229	0.0291	503	119	17
Leave-One-Out Sensitivity Analysis					
304122.2	0.0267	0.0299	490	112	16
288772.8	0.0350	0.0320	405	112	16
286053.9	0.0339	0.0604	459	112	16
266052.2	0.0402	0.0606	465	112	16
308453.0	0.0243	0.0296	489	112	16
308471.6	0.0243	0.0296	477	112	16
318579.6	0.0186	0.0294	457	112	16
341953.8	0.0055	0.0304	488	112	16
289098.6	0.0325	0.0624	492	112	16
238371.8	0.0429	0.0522	495	112	16
219798.9	0.0300	0.0447	461	112	16
304476.4	0.0265	0.0281	472	112	16
276152.2	0.0378	0.0588	481	112	16
300988.0	0.0285	0.0286	484	112	16
303717.4	0.0270	0.0284	477	112	16
303011.6	0.0274	0.0287	479	112	16
303582.1	0.0270	0.0286	477	112	16

Notes: Each row in this table corresponds to a separate regression using the same dependent variable: the firms' self-reported monthly sales. The model absorbs polygon and pre/post-truce timing fixed effects. Standard errors are clustered at the polygon level. The first row reports the full-sample. The second panel re-estimates the same specification while dropping one polygon at a time to assess sensitivity to influential geographic units. For each regression, the table reports the estimated treatment coefficient, the conventional clustered p -value, and the wild-cluster bootstrap p -value based on 9,999 Rademacher draws. The 'Firms' column reports the total number of establishments underlying the polygon-level aggregates used in each regression sample. The similarity of the estimates across exclusions indicates that the main result is not driven by any single polygon.

Figure B.13: Sectoral Growth and Economic Proportions During the Truce



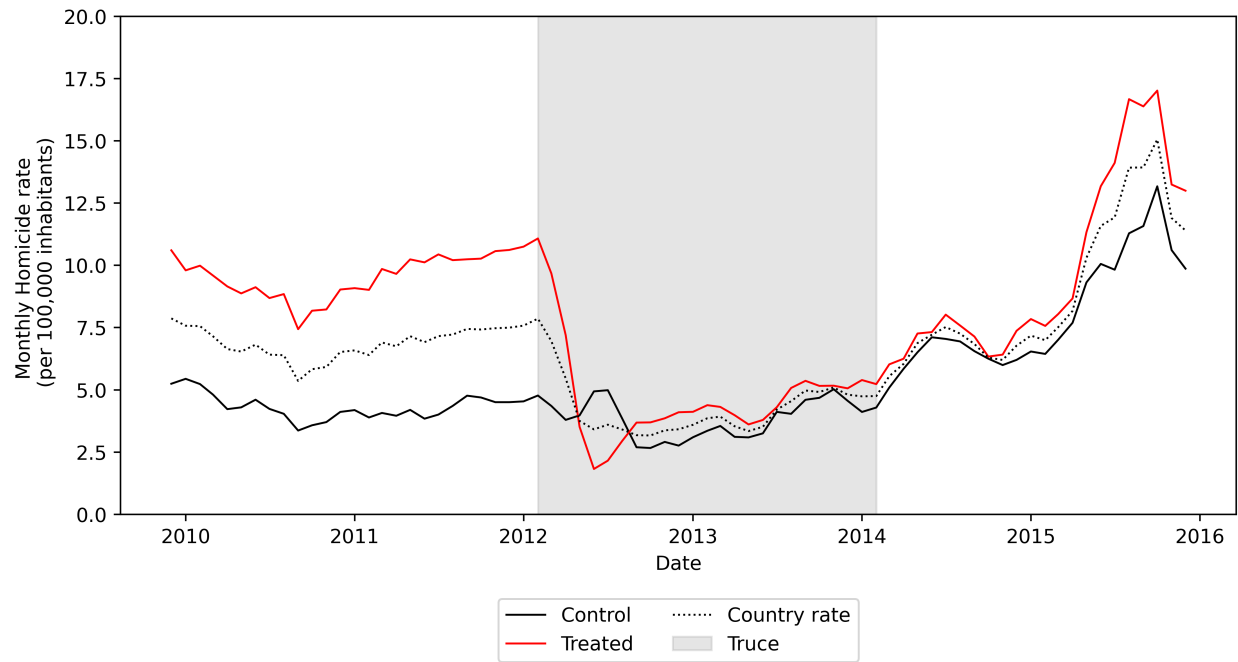
Notes: This graph presents a three-part analysis based on different sectors as classified by the CIIU Rev. 3 standard. The first column represents baseline regression estimates, calculated using sector-specific outcome variables to assess the impact of the truce on each sector. The second column indicates the relative growth in each sector due to the truce, calculated as the treatment effect estimator divided by the pre-truce levels of the corresponding indicator. The third column provides context by showing the percentage that each sector contributes to the overall economy just prior to the truce. The sample exclude top 5% firms.

Table B.8: Spillover Effects defining Treatment Based on Declines in Homicide Rates During the Truce Period

Variables	Workers	Workers	Workers	Number of firms	Number of firms	Number of firms
	Coef/SE	Coef/SE	Coef/SE	Coef/SE	Coef/SE	Coef/SE
Treatment	8.313*** (2.537)	7.609*** (2.530)	4.526** (2.218)	0.566*** (0.184)	0.493*** (0.187)	0.249 (0.182)
Spillover - neighbor		4.505*** (1.128)			0.466*** (0.108)	
Spillover - distance (km)			2.620*** (0.509)			0.219*** (0.041)
Obs.	25296	25296	25296	25296	25296	25296

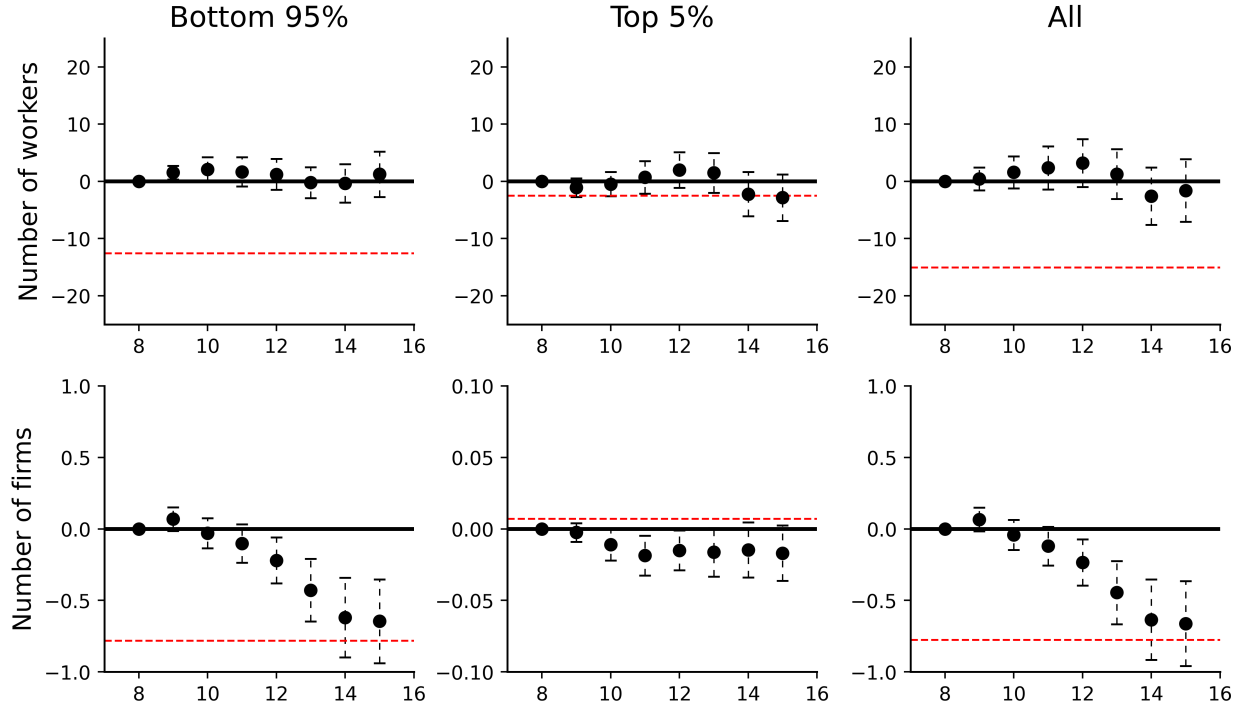
Notes: This table reports difference-in-differences estimates of spillover effects using the main treatment definition based on abnormal declines in homicide rates during the truce period. The specification extends Equation (1) by incorporating additional variables to capture indirect exposure to treated units. The variable Spillover – neighbor is a binary indicator equal to one if a unit shares a border with at least one treated unit. The variable Spillover – distance (km) measures the straight-line distance from each unit to the nearest treated unit. All regressions include unit and time fixed effects, and standard errors are clustered at the geographic unit level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure B.14: Homicides rates in Treated Areas and Control Areas after truce collapsed



Note: This graph illustrates the monthly homicide rate per 100,000 inhabitants, using a moving average. Additionally, the graph features a shaded gray area to delineate the period of the truce between gangs.

Figure B.15: Trends in the Number of Workers and Firms Post-Truce



Notes: This figure presents event-study estimates following the end of the gang truce, using a specification analogous to Equation (1). The sample includes treated units—defined as those with above-median homicide reductions during the truce period—and outcomes are shown separately for small and medium firms (left panel), large firms (center panel), and all firms combined (right panel). Time is aggregated into three-month intervals spanning 24 months after the truce. The red dotted horizontal line represents the average level of the outcome variable immediately prior the truce. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic unit level.

Figure B.16: Crime, Public Perceptions, and Consumer Confidence During the Truce and Non-Aggression Pact Periods



Notes: This figure shows descriptive evidence related to the 2012 gang truce and subsequent non-aggression pact periods in El Salvador. **Panel A** displays the national monthly homicide rate (per 100,000 inhabitants), aggregated from administrative data and smoothed with a 3-month moving average. **Panel B** illustrates the distribution of survey responses to a question on the most pressing problems facing the country, using LAPOP nationally representative household data. **Panel C** shows the evolution of the Consumer Confidence Index. The shaded gray areas indicate two key episodes: the gang truce (March 2012) and the non-aggression pact (April 2016).

B.3 Theoretical Framework

This appendix develops a parsimonious model that organizes the mechanisms suggested by the data. The economy is static, competitive, and features two geographic zones $z \in \{H, L\}$ producing a local non-tradable. For clarity in the exposition, whenever referring to the *low-crime zone* in the text I write *zone L*; symbols with subscript L (e.g., q_L, P_L) remain as standard notation.

The setup features two deviations from a standard local-demand framework: (i) crime enters the *consumer side* as a fear-cost wedge that raises the effective price of purchasing locally in the high-crime zone H ; and (ii) crime also shifts the *labor-supply side* by lowering participation at a given wage (crime-sensitive participation margin).

B.3.1 Environment and Agents

Time is static. There is a representative household of unit mass and a competitive sector of firms in each zone. Labor is perfectly mobile across zones and the production technology is linear in labor with common productivity $A > 0$.

Zones and crime. I index zones by $z \in \{H, L\}$ and let $c \geq 0$ denote the (exogenous) crime level specific to the high-crime zone H . For parsimony I normalize crime in the low-crime zone L to zero; allowing $c_L > 0$ is a straightforward variation.

B.3.2 Firms and Technology

Firms in each zone hire local labor ℓ_z and produce local output y_z using

$$y_z = A \ell_z, \quad A > 0.$$

Perfect competition and free mobility of labor imply that the (common) producer price equals unit cost,

$$p = \frac{w}{A},$$

where w is the common market wage. Local goods are non-tradable, so market clearing in each zone requires $y_z = q_z$, equivalently $A \ell_z = q_z$.

B.3.3 Goods, Crime, and Effective Prices (numéraire iceberg; household closure)

Crime affects consumption in H via a reduced-form fear-cost wedge $\chi(c)$ that raises the *effective consumer price* paid by households:

$$P_L \equiv p, \quad P_H \equiv p + \chi(c), \quad \chi(c) = \beta c, \quad \beta > 0.$$

Household income, utility, and budget. Preferences are quasilinear in the numéraire:

$$U = a_H \ln q_H + a_L \ln q_L + m, \quad a_H, a_L > 0.$$

Total numéraire income is

$$Y \equiv \bar{Y} + wL, \quad \bar{Y} \geq 0,$$

where $\bar{Y}$ is an exogenous endowment/transfer and wL is labor income (firms earn zero profits under perfect competition and $y_z = A\ell_z$). The representative household chooses (q_H, q_L, m) subject to

$$m + P_H q_H + P_L q_L = Y.$$

Resource accounting (numéraire sink). Of the expenditure $P_H q_H$, firms receive $p q_H$; the residual $\chi(c) q_H$ is numéraire *used and destroyed* (no agent receives it). Aggregating,

$$\bar{Y} + wL = m + p(q_H + q_L) + \chi(c) q_H,$$

and with $p = \frac{w}{A}$ and local clearing $q_z = A\ell_z$, we have $p(q_H + q_L) = wL$, so the accounting identity holds.

Feasibility (numéraire nonnegativity). Using $q_H^* = \frac{a_H}{p^* + \chi(c)} = \frac{a_H A}{w^* + A\beta c}$,

$$m^* = \bar{Y} - \chi(c) q_H^* = \bar{Y} - \beta c \cdot \frac{a_H A}{w^* + A\beta c}.$$

Thus the interior allocation is feasible provided

$$\boxed{\bar{Y} \geq \chi(c) q_H^* = \beta c \cdot \frac{a_H A}{w^* + A\beta c}}.$$

In what follows we assume $\bar{Y}$ is large enough so that $m^* \geq 0$. Because preferences are quasilinear, q_H and q_L depend only on (p, χ) ; m adjusts residually.

Demand. Utility maximization yields the Marshallian demands used throughout:

$$q_H = \frac{a_H}{p + \chi(c)}, \quad q_L = \frac{a_L}{p}.$$

Remark on “closed economy.” “Closed” here refers to the local non-tradable good and internal wage/price determination: goods and labor markets clear within the economy. The numéraire is modeled as an endowment/transfer $\bar{Y}$ financing residual consumption m .

B.3.4 Crime-Sensitive Supply, Labor Demand, and Interior Branch

Labor Supply. Crime also affects participation. I model this with a linear shift in the labor-supply schedule:

$$L^S(w, c) = s[w - \psi(c)]_+, \quad \psi(c) \equiv \gamma c, \quad s > 0, \gamma > 0,$$

where $[x]_+ = \max\{x, 0\}$. Parameter s is the slope (extensive-margin elasticity) of labor supply and γ governs how much crime lowers willingness to work (commuting risk, time losses, safety concerns).

Labor demand. Using $A\ell_z = q_z$, $p = w/A$, and $\chi(c) = \beta c$, the zone-level and aggregate labor demands are

$$\ell_H(w, c) = \frac{a_H}{w + A\beta c}, \quad \ell_L(w) = \frac{a_L}{w}, \quad L^D(w, c; a_H, a_L) = \frac{a_H}{w + A\beta c} + \frac{a_L}{w}.$$

Interior branch. For any $w \in (0, \gamma c]$, $L^S(w, c) = 0$ while $L^D(w, c; a_H, a_L) > 0$. At the participation boundary, $L^D(\gamma c, c) \in (0, \infty)$ if $c > 0$; and for any $c \geq 0$, $\lim_{w \downarrow 0} L^D(w, c) = +\infty$. Hence the market-clearing condition $L^S = L^D$ cannot hold at or below the participation threshold, and any equilibrium must satisfy $w^* > \gamma c$. On the interior domain $(\gamma c, \infty)$, $L^S(w, c) = s(w - \gamma c)$ is differentiable and strictly increasing, which I use in the existence and uniqueness proof below.

B.4 Competitive equilibrium: definition, existence, and uniqueness

I now characterize the competitive equilibrium for the baseline specification with size-only local demand and crime-sensitive labor supply introduced above.

B.4.1 Definition of equilibrium

Given parameters $(A, a_H, a_L, s, \beta, \gamma)$ and the crime level $c \geq 0$, a (competitive) equilibrium is a wage $w^* > 0$ and associated allocations

$$\begin{aligned} p^* &= \frac{w^*}{A}, & P_H^* &= p^* + \beta c, & P_L^* &= p^*, \\ q_H^* &= \frac{a_H}{P_H^*} = \frac{a_H A}{w^* + A\beta c}, & q_L^* &= \frac{a_L}{P_L^*} = \frac{a_L A}{w^*}, \\ \ell_H^* &= \frac{q_H^*}{A} = \frac{a_H}{w^* + A\beta c}, & \ell_L^* &= \frac{q_L^*}{A} = \frac{a_L}{w^*}, \end{aligned}$$

such that local non-tradable markets clear ($y_z^* = q_z^*$) and aggregate labor supply equals aggregate labor demand:

$$\begin{aligned} L^S(w^*, c) &= L^D(w^*, c; a_H, a_L), \\ L^S(w, c) &= s[w - \gamma c]_+, & L^D(w, c; a_H, a_L) &= \frac{a_H}{w + A\beta c} + \frac{a_L}{w}. \end{aligned} \tag{E}$$

I focus on the empirically relevant interior case $w^* > \gamma c$, under which $L^S(w, c) = s(w - \gamma c)$ is differentiable.

B.4.2 Existence and uniqueness (baseline size-only)

Lemma B.4.1 (Properties and boundary behavior of L^D) *For any $a_H, a_L, A > 0, \beta > 0$, and finite $c \geq 0$, the aggregate labor demand*

$$L^D(w, c; a_H, a_L) = \frac{a_H}{w + A\beta c} + \frac{a_L}{w}$$

is continuous on $(0, \infty)$, strictly decreasing in w , and satisfies

$$\lim_{w \downarrow 0} L^D(w, c) = +\infty, \quad \lim_{w \uparrow \infty} L^D(w, c) = 0, \quad \frac{\partial L^D}{\partial w}(w, c) = -\frac{a_H}{(w + A\beta c)^2} - \frac{a_L}{w^2} < 0.$$

Moreover, at the participation boundary $w = \gamma c$:

$$\begin{cases} L^D(\gamma c, c) = \frac{a_H}{(\gamma + A\beta)c} + \frac{a_L}{\gamma c} \in (0, \infty), & \text{if } c > 0, \\ \lim_{w \downarrow 0} L^D(w, 0) = +\infty, & \text{if } c = 0. \end{cases}$$

Proof. All statements follow directly from the closed form and elementary limits. ■

Lemma B.4.2 (Properties of L^S on the interior branch) For $w > \gamma c$, $L^S(w, c) = s(w - \gamma c)$ is continuous and strictly increasing in w , with $L^S(\gamma c, c) = 0$ and $\lim_{w \uparrow \infty} L^S(w, c) = +\infty$.

Proof. Immediate from linearity with $s > 0$. ■

Proposition 1 (Existence and uniqueness) Fix $a_H, a_L, A, s > 0$, $\beta, \gamma > 0$, and finite $c \geq 0$. There exists a unique $w^* > \gamma c$ solving (E).

Proof. Let $H(w; c) \equiv L^D(w, c; a_H, a_L) - L^S(w, c)$ for $w \geq \gamma c$.

Case $c > 0$. At $w = \gamma c$,

$$L^S(\gamma c, c) = 0, \quad L^D(\gamma c, c) = \frac{a_H}{(\gamma + A\beta)c} + \frac{a_L}{\gamma c} > 0,$$

so $H(\gamma c; c) > 0$.

Case $c = 0$. For $w > 0$, $H(w; 0)$ is continuous and

$$\lim_{w \downarrow 0} H(w; 0) = \lim_{w \downarrow 0} \left(\frac{a_H}{w} + \frac{a_L}{w} - s w \right) = +\infty,$$

since $L^D(w, 0) = (a_H + a_L)/w \rightarrow +\infty$ and $L^S(w, 0) = s w \rightarrow 0$.

In both cases H is continuous on $(\gamma c, \infty)$ and

$$\lim_{w \rightarrow \infty} H(w; c) = \lim_{w \rightarrow \infty} (L^D(w, c) - L^S(w, c)) = 0 - (+\infty) = -\infty.$$

By the Intermediate Value Theorem, there exists $w^* > \gamma c$ with $H(w^*; c) = 0$ (existence). Uniqueness follows because $L^D(\cdot, c)$ is strictly decreasing and $L^S(\cdot, c)$ is strictly increasing on $(\gamma c, \infty)$, so the two schedules can cross at most once. ■

Interiority and feasibility. Proposition 1 implies $w^* > \gamma c$, i.e., the interior branch is the relevant equilibrium (the kink is not binding). Given quasilinear preferences, the Marshallian quantities are income-independent; I assume the numéraire endowment Y is sufficiently large so that the implied $m^* \geq 0$ (feasibility of the interior allocation).

Equilibrium objects (summary). The unique equilibrium is characterized by the scalar w^* that solves (E). All remaining prices and allocations follow from

$$p^* = \frac{w^*}{A}, \quad P_H^* = p^* + \beta c, \quad P_L^* = p^*, \quad (q_H^*, q_L^*) = \left(\frac{a_H A}{w^* + A\beta c}, \frac{a_L A}{w^*} \right), \quad (\ell_H^*, \ell_L^*) = \left(\frac{a_H}{w^* + A\beta c}, \frac{a_L}{w^*} \right).$$

These expressions will be used below to derive the comparative statics with respect to crime.

B.5 Crime and equilibrium outcomes: wage and aggregate employment

We now study how the competitive equilibrium reacts to changes in crime c in the high-crime zone H , focusing on the equilibrium wage $w^*(c)$ and aggregate employment $L^*(c)$. Throughout we retain the size-only demand and crime-sensitive labor supply of (E), and the interior branch $w^* > \gamma c$.

B.5.1 Setup for comparative statics

Let

$$F(w, c) \equiv s[w - \gamma c] - \frac{a_H}{w + A\beta c} - \frac{a_L}{w},$$

so that the equilibrium condition is $F(w^*(c), c) = 0$. Define the convenient shorthands

$$X \equiv w^*(c) + A\beta c, \quad Z \equiv w^*(c).$$

Direct differentiation yields

$$F_w = s + \frac{a_H}{X^2} + \frac{a_L}{Z^2} > 0, \quad F_c = -s\gamma + \frac{a_H A\beta}{X^2}.$$

By the Implicit Function Theorem,

$$\frac{dw^*}{dc} = -\frac{F_c}{F_w} = -\frac{-s\gamma + \frac{a_H A\beta}{X^2}}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}}. \quad (9)$$

B.5.2 Equilibrium wage

Proposition 2 (Wage response to crime) *Under the baseline specification, the equilibrium wage is (locally) decreasing in crime if and only if*

$$\frac{a_H A \beta}{X^2} > s \gamma. \quad (C1)$$

Equivalently, when (C1) holds, a reduction in crime ($\Delta c < 0$) raises the wage ($\Delta w^ > 0$).*

Interpretation. Condition (C1) compares the *demand-side* impact of crime (the fear-cost wedge in H , scaled by its market size a_H and productivity A) to the *supply-side* inward shift (crime-sensitive participation $s\gamma$). If the demand effect dominates, crime reductions push the wage up; otherwise the wage response can be attenuated or (locally) flat/negative.

B.5.3 Aggregate employment

Aggregate employment equals

$$L^*(c) = L^S(w^*(c), c) = s[w^*(c) - \gamma c].$$

Differentiating and using (9) gives

$$\begin{aligned} \frac{dL^*}{dc} &= s \left(\frac{dw^*}{dc} - \gamma \right) = s \frac{-\left(-s\gamma + \frac{a_H A \beta}{X^2}\right) - \gamma \left(s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}\right)}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}} \\ &= -s \frac{a_H \frac{A\beta + \gamma}{X^2} + a_L \frac{\gamma}{Z^2}}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}}. \end{aligned}$$

Proposition 3 (Crime strictly lowers aggregate employment) *Fix $a_H, a_L, s, A > 0$ with $\beta > 0$ and $\gamma > 0$. Then*

$$\frac{dL^*}{dc} < 0.$$

Equivalently, any reduction in crime ($\Delta c < 0$) strictly increases aggregate employment ($\Delta L^ > 0$).*

Intuition. A crime reduction shifts *demand* outward by lowering the effective price in H and shifts *supply* outward by raising participation. Even if the wage response is muted (e.g., supply expands strongly), the outward movements ensure that total employment rises unambiguously.

B.6 Zone-by-zone effects: substitution across zones vs. participation gains

We now decompose the employment response by zone when crime c changes in H . Recall (size-only demand, crime-sensitive supply):

$$\ell_H = \frac{a_H}{w + A\beta c}, \quad \ell_L = \frac{a_L}{w}, \quad L^* = s[w^*(c) - \gamma c],$$

and define the shorthands

$$X \equiv w^* + A\beta c, \quad Z \equiv w^*.$$

Local derivatives and a convenient decomposition. Total derivatives with respect to c can be written as

$$\frac{d\ell_H}{dc} = \underbrace{\left(-\frac{a_H A\beta}{X^2}\right)}_{\text{direct demand (crime wedge in } H) \text{ at fixed } w} + \underbrace{\left(-\frac{a_H}{X^2}\right) \frac{dw^*}{dc}}_{\text{wage (general-equilibrium) channel}}, \quad (10)$$

$$\frac{d\ell_L}{dc} = \underbrace{0}_{\text{no direct crime term in the low-crime zone}} + \underbrace{\left(-\frac{a_L}{Z^2}\right) \frac{dw^*}{dc}}_{\text{wage (general-equilibrium) channel}}. \quad (11)$$

Thus, any change in employment in the low-crime zone is *purely* a reallocation (via the wage), while the high-crime zone combines a direct local-demand force with the wage channel.

Using the equilibrium condition (Section B.4), the wage derivative is

$$\frac{dw^*}{dc} = -\frac{-s\gamma + \frac{a_H A\beta}{X^2}}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}} \equiv -\frac{F_c}{F_w}, \quad F_w > 0. \quad (12)$$

B.6.1 When does cross-zone substitution occur?

A *crime reduction* ($\Delta c < 0$) raises aggregate employment L^* (Proposition 3). Whether we also observe reallocation away from the low-crime zone L hinges on the wage response.

Low-crime zone monotonicity. Employment in the low-crime zone is $\ell_L(w) = a_L/w$, which is strictly decreasing in w . Thus, for any finite change from c to c' with $w_1 = w^*(c) > 0$ and $w_2 = w^*(c') > 0$,

$$\Delta \ell_L = \ell_L(w_2) - \ell_L(w_1) = a_L \left(\frac{1}{w_2} - \frac{1}{w_1} \right) = -a_L \frac{\Delta w^*}{w_1 w_2}, \quad \Delta w^* \equiv w_2 - w_1,$$

so

$$\text{sign}(\Delta \ell_L) = -\text{sign}(\Delta w^*).$$

Local vs. finite changes. At the margin, reallocation from L occurs iff the wage increases when crime falls:

$$dc < 0 \Rightarrow dw^* > 0 \iff d\ell_L < 0.$$

In our baseline, this local wage increase holds *iff* condition

$$\frac{a_H A \beta}{X^2} > s \gamma \iff \frac{dw^*}{dc} < 0, \quad (\text{C1})$$

with $X \equiv w^* + A\beta c$. For a *finite* crime change Δc , the global implication $\Delta c < 0 \Rightarrow \Delta w^* > 0 \Rightarrow \Delta \ell_L < 0$ follows provided the sign $dw^*/dc < 0$ is maintained along the entire path in c (equivalently, (C1) holds for all intermediate crime levels).

Why ℓ_H must rise. Adding up $L^* = \ell_H + \ell_L$ and using $\Delta L^* > 0$:

- If (C1) holds, then $\Delta \ell_L < 0$ and thus

$$\Delta \ell_H = \Delta L^* - \Delta \ell_L > 0.$$

- If (C1) fails, the wage is weakly non-increasing, so the direct reduction in the H -wedge (lower $A\beta c$ in $w^* + A\beta c$) together with a flat/declining wage implies $\Delta \ell_H > 0$.

Hence, with a crime reduction, employment in H rises unambiguously; whether employment in L falls depends solely on the sign of the wage response via (C1).

Formal proof. Let $X \equiv w^* + A\beta c$ and $Z \equiv w^*$. Zone- H employment is

$$\ell_H = \frac{a_H}{X} \implies \frac{d\ell_H}{dc} = -\frac{a_H}{X^2} \left(\frac{dw^*}{dc} + A\beta \right).$$

Using the equilibrium condition $F(w, c) = 0$ with

$$F_w = s + \frac{a_H}{X^2} + \frac{a_L}{Z^2} > 0, \quad F_c = -s\gamma + \frac{a_H A \beta}{X^2},$$

the Implicit Function Theorem gives

$$\frac{dw^*}{dc} = -\frac{F_c}{F_w} \Rightarrow \frac{dw^*}{dc} + A\beta = \frac{-F_c + A\beta F_w}{F_w} = \frac{s(\gamma + A\beta) + A\beta \frac{a_L}{Z^2}}{F_w} > 0.$$

Therefore $\frac{d\ell_H}{dc} < 0$, i.e., a reduction in crime ($\Delta c < 0$) *strictly* increases employment in H :

$$\Delta c < 0 \Rightarrow \Delta \ell_H > 0.$$

B.6.2 An intuitive reallocation index

To keep the interpretation simple, summarize the composition of the employment gain from a crime reduction ($\Delta c < 0$) with a single *intensity index*:

$$\Delta L^* = \Delta \ell_H + \Delta \ell_L, \quad \text{with } \Delta L^* > 0 \text{ under } \beta, \gamma > 0.$$

Definition (reallocation intensity).

Reallocation index $I_{\text{sub}} \equiv \frac{\max\{0, -\Delta \ell_L\}}{\Delta L^*}.$

When is reallocation active? Reallocation is active ($I_{\text{sub}} > 0$) if and only if the wage rises with the crime reduction, which occurs under the dominance condition

$$\frac{a_H A \beta}{X^2} > s \gamma \iff \Delta w^* > 0 \iff \Delta \ell_L < 0. \quad (\text{C1})$$

How to read the index.

- **Large I_{sub} (reallocation-intensive):** many jobs leave L relative to the net gain. Typical when a_H is large, the fear-cost channel β is strong (or $X = w^* + A\beta c$ is low), and supply sensitivity $s\gamma$ is modest. Empirically: wage up, ℓ_L down.
- **Small I_{sub} (participation-driven):** few jobs leave L compared to the net gain. Typical when s and/or γ are high (wage moves little or falls), or when the base in L is small (a_L small / $Z = w^*$ large). Empirically: wage flat/down, ℓ_L flat/up.